UNIT - I : INTRODUCTION

Soil formation | Types | Weathering | Transport & Deposition | Classification

Subject: 23A01703b Subsurface Investigation and Instrumentation
Why this unit? Before we investigate any soil, we must know how it was formed - because the origin tells us its layering, grain size, strength and likely problems.

1. What is soil?

Soil is the unconsolidated natural collection of mineral grains, organic matter, water and air that lies above bed rock. To a civil engineer it is the material that (i) supports foundations, (ii) is used as a construction material (dams, embankments, roads) and (iii) must be excavated or retained.

- Geologist / Agriculturist: soil = top weathered layer that supports plant life.
- Civil engineer: soil = any loose, uncemented deposit that can be separated by light agitation in water (e.g. gravel, sand, silt, clay).
- Rock: a hard, cemented mass of minerals that is not broken down by simple agitation in water.

Three-phase nature of soil

Soil = solids (mineral grains) + water (in voids) + air (in voids). If all voids are full of water it is saturated; if no water it is dry; otherwise partly saturated. Properties such as strength and compressibility change with the amount of water, which is why we always record the water table during exploration.

2. Soil formation (Pedogenesis)

All soils are formed from parent rock by the process of weathering (breaking down) followed in many cases by transportation and deposition. Soil formation depends on five factors, remembered as C-O-R-P-T:

Factor	Effect on soil formation
Climate	Rainfall and temperature control the rate of weathering; hot-humid -> deep chemical weathering (laterite)
Organisms	Plants, bacteria, worms add humus, roots break rock
Relief (topography)	Steep slope -> erosion, thin soil; flat -> thick deposits
Parent material	Mineral composition of rock decides soil type (granite -> sandy, basalt -> clayey)
Time	Longer time -> deeper, more mature profile

Rock cycle - the big picture

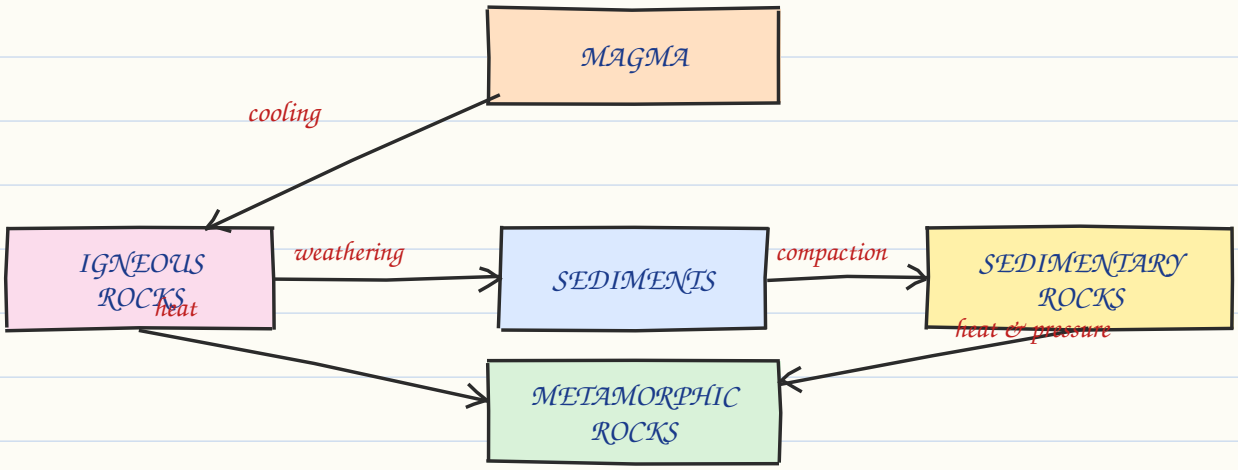


Fig 1.1 Rock cycle

Three families of rocks

Type	Formation	Examples
Igneous	Cooling and solidification of magma/lava. (Intrusive = granite, coarse; Extrusive = basalt, fine)	Granite, Basalt, Dolerite, Gabbro
Sedimentary	Deposition and cementation of sediments in layers (strata)	Sandstone, Limestone, Shale, Conglomerate
Metamorphic	Existing rocks altered by heat, pressure, chemically active fluids	Gneiss, Schist, Slate, Marble, Quartzite

Remember: Granite -> (metamorphism) -> Gneiss | Limestone -> Marble | Shale -> Slate | Sandstone -> Quartzite.

3. Weathering

Weathering is the in-situ disintegration and decomposition of rock at or near the earth surface by the action of atmosphere, water, temperature and living things. It does not involve movement; moving of the broken material is called erosion/transport.



Fig 1.2 Classification of weathering

3.1 Physical (mechanical) weathering

Rock breaks into smaller pieces *without change* in chemical composition. Produces gravel, sand and some silt - *coarse-grained, cohesionless* soils.

- *Temperature change (thermal expansion):* daily heating/cooling in deserts - different minerals expand unequally -> cracks.
- *Exfoliation:* outer sheets of rock peel off like onion skin due to heating/cooling and unloading.
- *Frost action:* water enters cracks, freezes, expands by about 9 % -> wedges rock apart.
- *Abrasion:* rubbing by moving water, wind-blown sand, glaciers.
- *Plant & animal action:* roots grow in cracks; burrowing animals.
- *Unloading:* removal of overburden releases pressure -> sheet joints.
- *Crystallisation of salts* in pores of rock in dry regions.

3.2 Chemical weathering

Minerals are chemically changed into new minerals (clay minerals) - produces fine-grained cohesive soils (silt, clay). Requires water; fastest in hot and humid climate.

Process	What happens	Example
Hydration	Minerals absorb water, swell	Anhydrite -> Gypsum
Hydrolysis	Water + mineral -> acid/base reaction	Feldspar -> Kaolinite (clay)
Oxidation	Reaction with oxygen (iron-bearing minerals) -> rust colour	Iron -> Hematite / Limonite
Carbonation	CO2 dissolved in rain (carbonic acid) attacks limestone	CaCO3 -> Ca(HCO3)2 soluble
Solution	Direct dissolving of soluble minerals	Rock salt, gypsum
Leaching	Soluble salts washed out of upper layers	Laterite formation

Key difference: Physical weathering -> particle SIZE changes. Chemical weathering -> composition changes -> CLAY minerals (kaolinite, illite, montmorillonite) appear. Clay minerals decide swelling, plasticity and cohesion.

Soil profile (horizons)

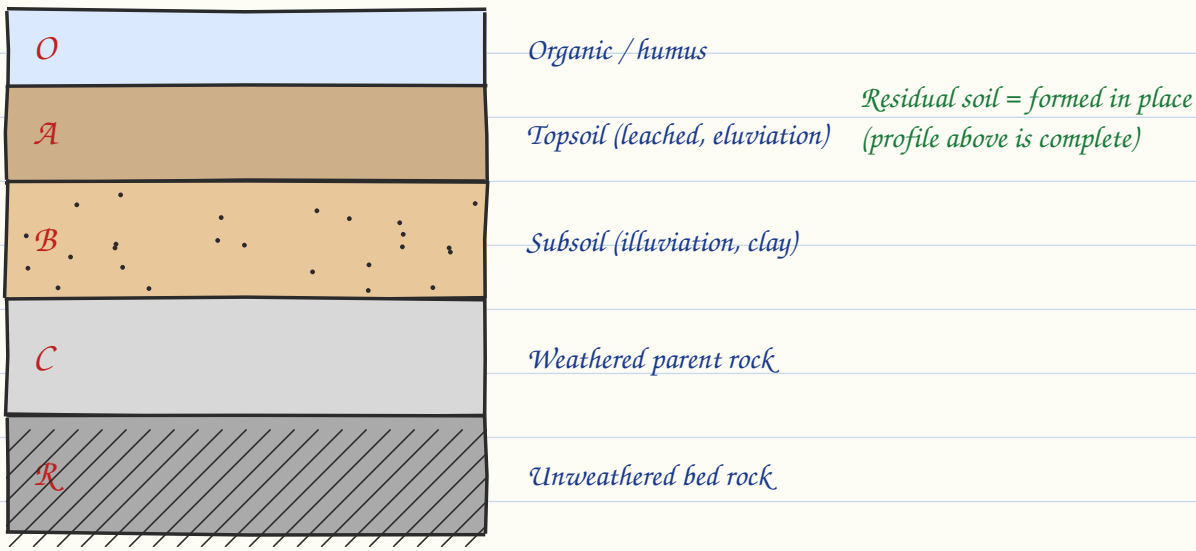


Fig 1.3 Soil profile of a residual soil

Soil moves downwards from the surface: leaching of fine particles from A (eluviation) and their accumulation in B (illuviation). Engineers mostly deal with B and C horizons.

4. Types of soils (by origin)

4.1 Residual soils

Weathered products that stay at the place of formation (rate of weathering > rate of removal). Particle sizes vary from fine at top to coarse and then rock at depth. Grains are angular, unsorted, no stratification. Strength increases with depth. Example: laterite of Kerala, black cotton soil over basalt.

4.2 Transported soils

Weathered products carried away and deposited elsewhere by wind, water, ice or gravity. They are stratified, grains are rounded and sorted. Classified by agency of transport:

Soil	Agency	Characteristics
Alluvial	River water	Rounded grains, well-stratified; sand lenses and clay; river plains/deltas (Ganga, Godavari, Krishna)
Lacustrine	Lake deposits	Very fine, silt and clay, varved (seasonal layers)
Marine	Sea water	Soft, very compressible clays; coastal; sensitive; salt content
Aeolian	Wind	Uniform fine sand (dunes) or silt (loess); loose, collapses on wetting
Glacial	Ice	Boulder clay / till - mixture of all sizes, unsorted
Colluvial	Gravity (landslide, creep)	Heterogeneous, angular, at foot of slopes - talus

4.3 Special Indian soils (important)

Soil	Features	Engineering problem
Black Cotton Soil (Regur)	Derived from basalt; contains MONTMORILLONITE; found in A.P., Maharashtra, M.P., Gujarat	High swelling and shrinkage, deep cracks in summer; poor for foundations/roads

Soil	Features	Engineering problem
Laterite	Hot humid, leaching of silica; rich in iron/aluminium oxides; red-brown	Hardens on exposure; good as road material
Alluvial	Ganga-Brahmaputra plains, deltas	Generally fertile, moderate strength
Red soil	Weathered granite/gneiss; iron oxide; common in Anantapur, Rayalaseema	Low in lime; fairly good bearing, sandy-clayey
Desert soil	Rajasthan; wind-blown sand	Loose, low cohesion, wind erosion
Marine clay	Coastal Gujarat, Kerala, Chennai	Very soft; consolidation settlement

5. Soil transport, deposition and stratification

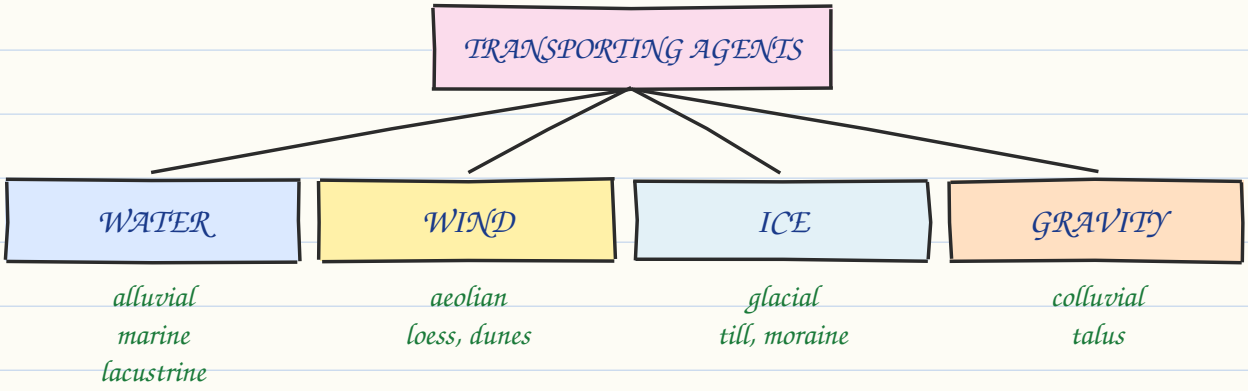


Fig 1.4 Agents of transportation

5.1 Transportation

- Water carries particles by suspension (silt, clay), saltation (bouncing sand) and traction (rolling gravel). Capacity is proportional to velocity^6 (approx.).
- Wind moves fine sand by saltation, dust by suspension over long distances.
- Glaciers carry everything, from boulders to rock flour.
- Gravity acts on slopes: rock fall, landslide, soil creep, mud flow.

During transport particles are sorted (heavier ones fall first), rounded (abrasion) and reduced in size. Hence particle shape and size give clues to the distance travelled.

5.2 Deposition

Deposition occurs when the carrying power of the agent drops - e.g. river entering a lake or sea (delta), river flattening (flood plain), wind losing speed behind obstacles (dune), melting of glacier (moraine).

- Coarse particles settle first (near hills), fine particles settle far away (lakes and sea).
- In still water clay particles flocculate in salt water (open, honeycomb structure) -> very high void ratio; in fresh water they form a dispersed structure.
- Deposits are often normally consolidated at the beginning and may later become over-consolidated due to erosion of overburden, glaciation or desiccation.

5.3 Stratification

Deposition in successive layers (strata/beds) over time is called stratification. Seasonal changes in flow give alternate layers of silt and clay called varves (lake clay). Strata may be horizontal, inclined, lenticular or may pinch out. For site investigation this means soil properties change quickly in the vertical direction and sometimes horizontally.

> Importance for investigation: Residual soil -> no stratification, properties vary gradually with depth. Transported soil -> well stratified, boreholes must be deep and closely spaced to catch lenses, pockets and thin weak layers (e.g. soft clay between sand).

6. Soil structure and clay minerals (short note)

- Single-grained (coarse soils - gravel/sand, gravity forces).
- Honeycomb (fine sand/silt, bridging).
- Flocculent (clays in salt water - edge-to-face attraction).
- Dispersed (clays in fresh water - face-to-face, parallel).

Clay mineral	Structure	Properties
Kaolinite	1:1, strong H-bond	Stable, low swelling, low plasticity
Illite	2:1, K+ bonding	Medium swelling
Montmorillonite	2:1, weak bonding by water	Very high swelling - black cotton soil

7. Soil classification

Classification places soils with similar engineering behaviour in the same group so that experience can be transferred and a common language is used in boring logs. The tests used are grain size analysis (sieve + hydrometer) and Atterberg limits (liquid limit wL, plastic limit wP, shrinkage limit ws).

7.1 Particle size classification (IS: 1498)

Fraction	Size range (mm)
Gravel (G)	80 - 4.75
Coarse sand	4.75 - 2.0
Medium sand	2.0 - 0.425
Fine sand	0.425 - 0.075
Silt (M)	0.075 - 0.002
Clay (C)	< 0.002

Cobbles: 300-80 mm; Boulders: > 300 mm. Total sand = 4.75 to 0.075 mm.

7.2 Atterberg limits & indices

- Liquid limit wL: water content at which soil changes from plastic to liquid (25 blows in Casagrande cup).
- Plastic limit wP: water content at which soil crumbles when rolled to 3 mm thread.
- Shrinkage limit ws: water content below which no further volume reduction occurs on drying.
- Plasticity index Ip = wL - wP | Liquidity index IL = (w - wP)/Ip | Consistency index Ic = (wL - w)/Ip | Activity A = Ip / (% clay).

7.3 Indian Standard Soil Classification System (ISSCS) - IS:1498-1970

It is similar to USCS (Casagrande). Step 1: if more than 50 % retained on 75 micron sieve -> coarse grained; otherwise fine grained.

Group symbol	Meaning
GW	Well graded gravel (Cu > 4, Cc 1-3)
GP	Poorly graded gravel
GM / GC	Silty gravel / clayey gravel
SW / SP	Well graded sand (Cu > 6) / poorly graded sand
SM / SC	Silty sand / clayey sand
ML, MI, MH	Silt - low, intermediate, high compressibility
CL, CI, CH	Clay - low, intermediate, high compressibility
OL, OI, OH	Organic silt / clay
Pt	Peat - highly organic

Gradation formulae: Cu = D60/D10 | Cc = (D30)^2 / (D10 x D60). Well graded gravel: Cu >= 4 and Cc 1 to 3. Well graded sand: Cu >= 6 and Cc 1 to 3. Otherwise poorly graded (P).

Plasticity chart (for fine soils)

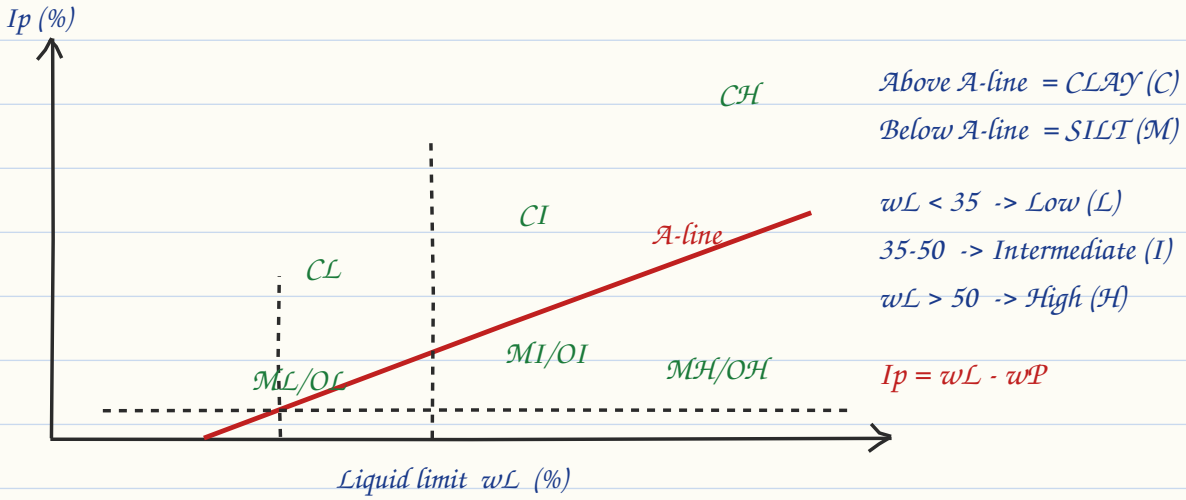


Fig 1.5 Plasticity chart (A-line: Ip = 0.73 (wL - 20))

7.4 Other systems

- **USCS (Unified):** same symbols as IS; liquid limit divides into L (< 50) and H (> 50) only.
- **AASHTO (Highway):** groups A-1 to A-7 using sieve analysis + wL + Ip + group index: GI = 0.2a + 0.005ac + 0.01bd. Lower GI -> better subgrade.
- **Textural (triangular) classification:** uses % sand-silt-clay (USDA) - mainly for agriculture.
- **Field classification:** dry strength, dilatancy, toughness tests (see Unit IV, visual identification).

7.5 Consistency of clays (field)

Consistency	Unconfined strength qu (kPa)	Field sign
Very soft	< 25	Squeezes between fingers
Soft	25 - 50	Moulded easily
Medium	50 - 100	Moulded with strong pressure
Stiff	100 - 200	Thumb dents with effort
Very stiff	200 - 400	Thumbnail indents
Hard	> 400	Thumbnail hardly marks

8. Worked examples

Example 1 (gradation). Sieve analysis gives D10 = 0.08 mm, D30 = 0.30 mm, D60 = 0.90 mm.

Cu = D60/D10 = 0.90/0.08 = **11.25** (> 6)

Cc = (0.30)^2 / (0.08 × 0.90) = 0.09/0.072 = **1.25** (between 1 and 3)

Both satisfied -> **well graded sand (SW)** if < 5 % fines.

Example 2 (consistency). A clay has wL = 55 %, wP = 25 %, natural water content w = 40 %.

Ip = 55 - 25 = 30 %

IL = (40 - 25)/30 = 0.50 ; Ic = (55 - 40)/30 = 0.50 (stiff to medium clay)

A-line check: 0.73 × (55 - 20) = 25.6 < 30 -> above A-line, wL > 50 -> CH (clay of high compressibility).

Example 3 (percentage). 500 g dry soil: retained on 4.75 mm = 40 g, passing 4.75 & retained 75 micron = 330 g, passing 75 micron = 130 g.

Gravel = 8 %, Sand = 66 %, Fines = 26 % -> coarse grained (fines < 50 %); fines > 12 % -> SM or SC (check Ip of fines).

Residual vs transported soil - quick comparison

Point	Residual soil	Transported soil
Place	Same place as parent rock	Moved by water/wind/ice/gravity
Grain shape	Angular	Rounded to sub-rounded
Sorting	Poor, varied sizes	Well sorted (in water/wind)
Stratification	Absent	Present, often lenticular
Variation	Gradual with depth	Abrupt vertical and lateral
Typical problem	Weathered rock pockets, collapse on wetting	Soft clay layers, liquefiable sand, settlement

Engineering significance of weathering

- Degree of weathering decides depth of foundation and the rock/soil boundary (rock head) that boring must locate.
- Chemical weathering creates clay minerals -> swelling/shrinking and low strength.
- Heavily weathered rock can behave like dense soil: use SPT N > 50 / refusal and core recovery to identify it.
- Frost and thermal cracks create joints in rock which govern permeability and stability of slopes and tunnels.

Memory tricks

- CORPT = Climate, Organisms, Relief, Parent material, Time.
- Particle sizes: G-S-M-C = Gravel > Sand > Silt (M) > Clay, boundaries 4.75, 0.075, 0.002 mm.
- "Above A-line = Clay, Below = Mud (silt)".
- Alluvial = water, Aeolian = air, Glacial = ice, Colluvial = collapse (gravity).

9. Important questions

- Explain the process of soil formation and the factors that control it. (long)
- Differentiate between physical and chemical weathering with examples.
- Explain the types of transported soils and their engineering characteristics.
- Write short notes on stratification and varved clay.
- Describe the IS soil classification system with the plasticity chart.
- A soil has wL = 62, wP = 28. Find Ip and classify (A-line: 0.73 × 42 = 30.7; Ip = 34 > 30.7 so CH).

Quick revision: Rock --weathering--> soil --transport--> deposit --burial--> sedimentary rock. Residual = in place, angular. Transported = stratified, rounded. Classification = grain size + plasticity.

UNIT - II : METHODS OF SOIL EXPLORATION

Boring | Auguring | Drilling | Samplers | Soil Sampling

Aim of exploration: (1) know the sequence and thickness of strata, (2) locate the ground water table, (3) collect samples for lab tests, (4) find the depth to bed rock, (5) get strength and compressibility data for design of foundation/earth structures.

1. Stages of a site investigation

- **Reconnaissance / desk study** - maps, aerial photos, geological records, nearby structures.
- **Preliminary exploration** - few boreholes, test pits to get general picture.
- **Detailed exploration** - number, depth and spacing of boreholes planned on the basis of the structure.
- **Testing & report** - field tests (Unit IV), lab tests, final report (Unit V).

Classification of exploration methods

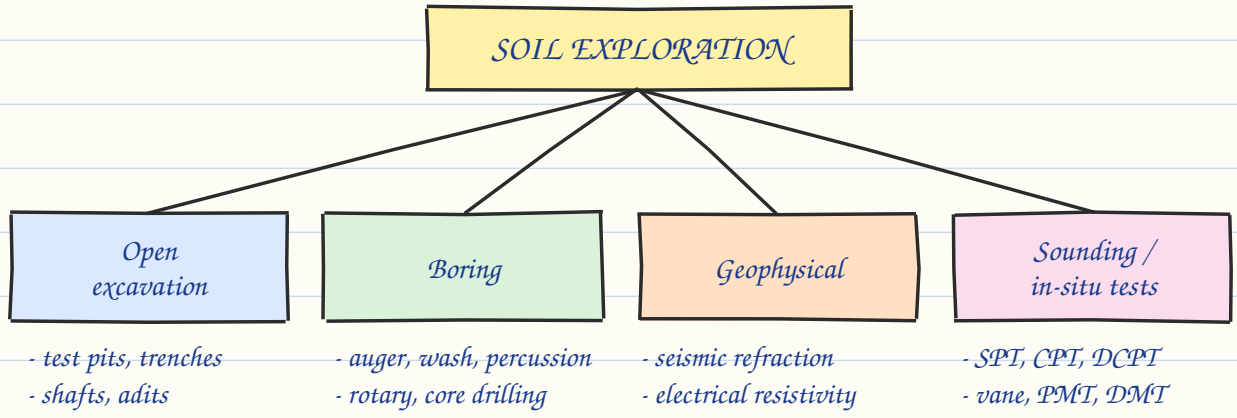


Fig 2.1 Methods of subsurface exploration

1.1 Open excavation methods

- *Test pits (trial pits):* 1.2 m x 1.2 m or more, used to depth 3 - 5 m (dry ground). Soil is seen in-situ; undisturbed block samples can be cut out. Cheap but limited depth and dangerous in loose/wet soil - support the sides.
- *Trenches:* long narrow excavation (1 m wide) to study continuity of strata, fault, pipelines.
- *Shafts:* deep excavation (> 3 m) used for tunnels, large dams and in rock; sides are timbered.
- *Adits / drifts:* horizontal tunnels in hill side - for dam abutments.

1.2 Depth and spacing of boreholes (IS:1892 guideline)

Structure	Spacing of boreholes	Depth of exploration
Isolated building	at least 4 (corners) + 1 at centre	1.5 x width of footing (or 1.5 B) below foundation; to 2 x B for heavy loads
Multi-storeyed buildings	10 - 30 m grid	Till strata of adequate bearing (significant stress < 10 % of qnet)
Highways	30 - 60 m (200 m on uniform ground)	2 - 3 m below subgrade
Earth dams / embankments	30 - 60 m along axis	Up to hard stratum, 2-3 times dam height
Bridges	At each pier/abutment	Up to 2 x B below pier or to rock (min. 3 m in rock)

Rule of thumb: explore to the depth where the increase in vertical stress due to foundation = 10 % (or 20 % in some codes) of the foundation pressure. Continue 3 m into bed rock if rock is met; in case of a floating raft explore below the base up to 1.5 times width.

2. Boring - making a hole in the ground

Boring means advancing a hole (borehole, diameter 50 - 200 mm) to explore the soil in depth. The method chosen depends on soil type, depth, ground water, equipment available and the purpose (sampling / testing / logging only).

Method	How it works	Suitable for
Auger boring	Soil is cut and lifted by auger turned by hand or machine	Soft to stiff cohesive soils above water table; depth up to 6 m (hand) or 30 m (power)
Wash boring	Casing driven; chopping bit on hollow rod, water jetted, cuttings flow up between rod and casing	Sand, silt, soft clays (not gravel/boulders); cheap; sample quality poor
Percussion boring	Heavy chisel / bailer / shell raised and dropped (cable tool)	All soils incl. gravel and soft rock; slow
Rotary drilling	Rotating bit with circulating drilling mud/water/air	Hard soils and rock; deep holes; high speed
Core drilling	Diamond / tungsten carbide bit with core barrel to cut rock cores	Rock; gives cores for RQD, strength tests

2.1 Wash boring (important)

- A casing (50 - 100 mm) is driven by hammer/driving head.
- A hollow drill rod with chopping bit is lowered inside; it is raised and dropped, and rotated by hand to cut the soil.
- Water is forced through the rod by a pump; it comes out of the bit, carries the cuttings up between rod and casing to a settling tank.
- Sample from the tank = disturbed, washed (fines lost). Change of colour/speed is noted to identify strata.
- To take samples, rod is withdrawn and a sampler is attached and driven in.

2.2 Percussion (cable tool) drilling

A heavy tool - chisel (for hard soil/rock), bailer or shell (to remove slurry) - is attached to a wire rope and repeatedly lifted & dropped from a tripod. Water is added to make slurry in dry holes. Good for boulders and gravels but disturbs the soil below the bottom of the hole; used widely for water wells.

2.3 Rotary drilling

- The bit is rotated under axial thrust by a rotary head; drilling fluid (bentonite mud, water or air-foam) is pumped through hollow rods to cool the bit, carry the cuttings and stabilise the borehole wall.
- Bits: drag (fishtail) bit for soft soils, roller (tricone) bit for medium rock, diamond/tungsten carbide bit for hard rock and coring.
- Fast and gives high productivity; hole diameter 50 - 150 mm. Used for depths more than 30 m.
- Core barrels: single tube, double tube (inner tube stays stationary, protects core), triple tube; sizes EX, AX, BX, NX (NX core dia = 54 mm).

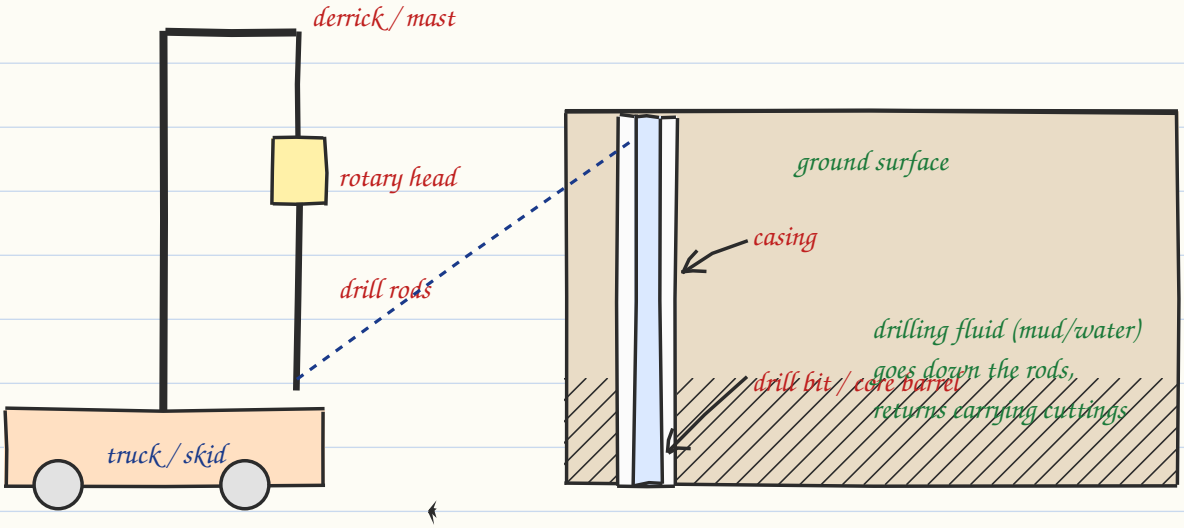


Fig 2.2 Rotary drilling rig with circulating fluid

3. Auguring

Auger boring is the simplest method; the auger is rotated into ground and soil is lifted on its flights. It is dry boring - used above ground water. In loose sand below water table the hole collapses, so casing is used.

3.1 Types of augers and their usage

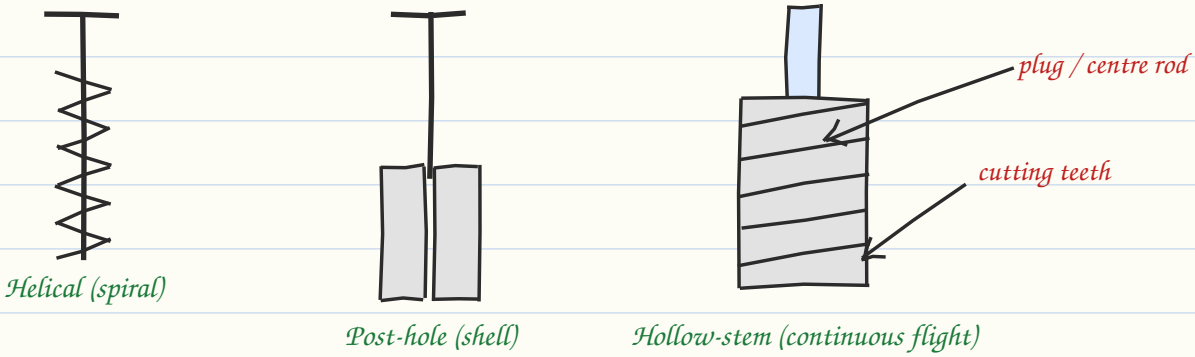


Fig 2.3 Common types of augers

Type	Description	Use
Helical (spiral) auger	Screw-like, 75 - 200 mm dia, hand operated with T-handle, extensions of 1 m	Stiff, dry, cohesive soil up to 6 - 10 m; shallow investigation, road projects
Post-hole (shell) auger	Two half-cylinders with cutting edges, like a bucket	Soft to medium clay, silt above W.T.; faster, brings up sample
Iwan (Dutch) auger	Open-ended cylindrical, 100 - 200 mm; hand rotated	Soft and sandy soils, shallow
Continuous flight (power) auger	Machine rotated long flights, soil comes up the flights continuously	Quick logging to 30 m; highways, shallow foundations
Hollow-stem auger	Flights wound on a hollow tube; plug removed to insert sampler/SPT through it	Soils with water table, caving sands - gives clean access for SPT & thin-wall sampling
Bucket (large dia) auger	Up to 1 m diameter bucket-like	Bored pile excavation, inspection of strata by person lowered

Advantages of auger boring: cheap, rapid, small equipment, no water needed.
Limitations: unsuitable below W.T. in sands, gravels or boulders; samples are disturbed and mixed; depth of strata change is only approximate.

Machinery used for drilling

- Hand operated: tripod, pulley, hand augers, hammers - small jobs, up to 10 m.
- Truck / skid mounted rotary rigs: engine + hydraulic feed + rotary table (top-drive), mast, mud pump, winch.
- Percussion (churn) rigs: tripod/derrick with winch & spudding beam.
- Wire-line drill for deep core drilling, sonic drills for continuous cores in difficult ground, dual-tube reverse circulation for gravel & boulders.
- Accessories: casing, drill rods (AW, BW), bits, shoes, mud pump, water tank, tripod & pulley, SPT hammer (63.5 kg), bailers, tremie, core boxes.

4. Soil sampling

The quality of design depends on the quality of samples. A sample must represent the soil as found in the field. Disturbance may be caused by the drilling method, sampler wall thickness, stress relief, handling and storage.

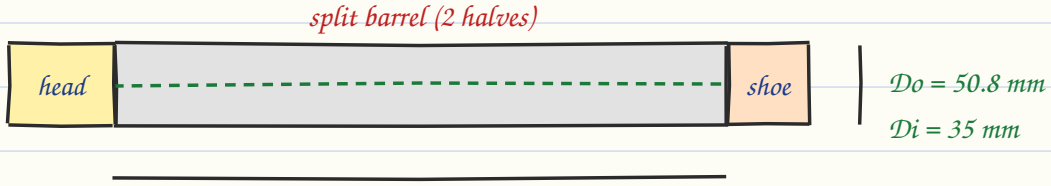
4.1 Types of samples

Sample	Description	Used for
Disturbed (non-representative)	Structure destroyed; composition may change (fines lost); collected from cuttings, auger, wash water	Visual identification only (logging)
Disturbed (representative)	Structure destroyed but all constituents and water content preserved (split spoon, auger cuttings sealed in jar)	Grain size, Atterberg limits, specific gravity, water content, compaction
Undisturbed	Natural structure, density & water content are preserved as far as practical (thin-wall tube, block, piston sample)	Shear strength, consolidation, permeability, unit weight
Block samples	Cut by hand from test pit or trench, waxed immediately	Best quality for clay, residual soils, sensitive/stiff soils
Core samples	Cylinder of rock from core barrel	RQD, compressive strength, rock tests

Truth about undisturbed samples: a truly undisturbed sample is impossible - stress relief, water change and sampling stresses always exist. So we say "undisturbed samples are those with minimum disturbance" (Hvorslev). Rule: design-quality sample = area ratio < 10 %, inside clearance 1 - 3 %, recovery ratio > 95 %.

4.2 Sampling methods / samplers

IS:2131 / ASTM D1586 standard split spoon

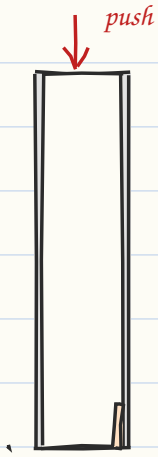


Ends in cutting shoe; ball valve in head vents water; sample retained by core-catcher.

Fig 2.4 Standard split-spoon sampler

- *Split-spoon (split barrel) sampler:* 50.8 mm OD, 34.9 mm ID, 457 mm long barrel; heavy walls -> disturbed but representative sample. Also used in SPT. Liner may be used.
- *Thin-walled (Shelby) tube sampler:* seamless steel tube, 50 - 100 mm OD, wall 1.25 - 3 mm, *pushed* (not hammered) steadily into soft to medium clays. Ends sealed with wax after taking out.
- *Piston sampler (stationary piston / Osterberg / Hvorslev):* thin tube with piston which keeps fluid pressure off the sample when it is pushed; for soft sensitive clays, recovery near 100 %.
- *Denison sampler:* double-tube rotating core barrel with inner non-rotating liner; for stiff clays and hard soils.
- *Pitcher sampler:* spring-loaded inner tube projects ahead of the outer rotating tube; for stiff to hard clays, and soft rock.
- *Open drive sampler:* simple open tube driven by hammer; *Foil / Swedish sampler:* for very soft clays - continuous samples.

- Scraper bucket sampler: for sampling gravelly, sandy soil below water.
- Rotary core barrels (NX etc.) with diamond bits: rocks.



Shelby tube (thin walled)

wall 1.25 - 3 mm steel / brass

length = 5 to 10 x diameter

Di = inner dia at cutting edge, De = Dc = outer dia

cutting edge is sharp (taper 6 - 10 deg)

Common sizes: 50 mm, 75 mm, 100 mm OD

Pushed (not driven) into soft/medium clay

Fig 2.5 Thin walled sampler

4.3 Sampler specifications - design parameters

Disturbance depends on the geometry of cutting edge and wall, defined by the following ratios (where Do = outer dia of tube at cutting edge, Di = inner diameter at cutting edge, Dc = inside dia of sampler tube above cutting edge, De = outside dia of tube):

Parameter	Formula	Permissible for undisturbed sampler
Area ratio Ar	(De^2 - Di^2)/Di^2 x 100	< 10 % (soft clay) ; 10-20 % (stiff clay / sand); split spoon about 100 %
Inside clearance Ci	(Dc - Di)/Di x 100	0.5 - 3 % (to reduce friction on the sample)
Outside clearance Co	(Do - De)/De x 100	0 - 2 % (to reduce driving effort)
Recovery ratio Lr	Length of sample recovered/length of tube pushed	> 0.96 for good sample
Length/diameter	L / D	Clay: 5 - 10 ; sand: up to 20

Example: Thin tube with De = 54 mm, Di = 50 mm: Ar = (54^2 - 50^2)/50^2 x 100 = (2916 - 2500)/2500 x 100 = 16.6 % (> 10 %, so only for stiff clay, moderate quality). For a split spoon De = 50.8, Di = 34.9: Ar = (2580.6 - 1218)/1218 = 112 % -> highly disturbed.

4.4 Sample sizes (usual)

Test	Minimum sample size
Visual / identification	500 g from each stratum (jar)
Grain size, Atterberg limits, G	1 - 2 kg disturbed soil
Compaction (Proctor)	25 - 30 kg bulk sample
Unconfined / triaxial test	38 mm dia x 76 mm length (min. 3 specimens) from 75 mm tube
Consolidation test	60 mm dia x 20 mm specimen (from 75 - 100 mm tube)
Rock compressive strength	NX cores, L/D = 2 : 1
Water sample	1 litre in clean bottle (chemical tests)

5. Handling, preservation, storage and transport of samples

- Label every sample immediately: project, borehole no., depth, date, sampler, sample no., top & bottom, method, water table. Keep a separate record.
- Disturbed samples: place in air-tight jars / polythene bags; add label inside and outside; protect from sun to keep natural moisture.
- Undisturbed samples: seal both ends of the tube with molten wax (3 layers, 25 mm thick) + cap; store vertically, top up. For block samples: wrap in cloth, dip in wax, box with sawdust.
- Storage: cool, shaded, humidity-controlled room; avoid freezing/vibration. Test within 2 - 3 weeks (soft clays within 1 week) to avoid water migration & ageing.

- Transport: use padded wooden boxes, cushion of foam/sawdust, handle with care - NO jolt or dropping; keep tubes vertical.
- Extrusion in the lab should be in the same direction as the sample entered the tube.
- Retain all rock cores in core boxes with depth marking; photograph them.

5.1 Sample preparation in laboratory

- Disturbed: air-dry (or oven-dry below 60 deg C for organic soils), break lumps with wooden mallet, pass through 4.75 mm sieve, use quartering (coning) to get representative portion.
- Undisturbed: extrude, cut specimen with wire saw, trim in lathe to required dia and length, keep covered with wet cloth to avoid drying.
- Moisture content of each specimen should be taken from the trimmings.

5.2 Factors causing disturbance of samples

- Method of boring: driving casing, jetting water below the casing tip, over-drilling or heave in the hole bottom (sand "boiling").
- Stress relief: removal of overburden pressure -> soil swells, structure loosens.
- Sampler design: thick wall, large area ratio, blunt cutting edge, no inside clearance -> shear distortion of edges of sample.
- Sampling operation: hammering instead of pushing, jerky push, leaving tube in hole (suction on withdrawal) - use slow steady push and rotate before pulling.
- Handling: vibration in transport, sun heat, loss of moisture, extrusion in wrong direction, long storage.

5.3 Quality classes of samples (Hvorslev)

Class	Quality	Typical properties obtainable
1	Undisturbed (block, piston)	Strength, stiffness, consolidation, permeability
2	Slightly disturbed (thin tube)	Water content, unit weight, strength (approx.)

Class	Quality	Typical properties obtainable
3	Remoulded - all constituents retained (split spoon)	Grading, Atterberg limits, water content, G
4	Mixed / washed fines lost	Only soil type, rough grading
5	Debris / cuttings	Strata boundary identification only

5.4 Choice of drilling method - comparison

Method	Merits	Demerits
Auger	Cheap, quick, simple	Limited depth; fails below W.T. in sand/gravel
Wash boring	Low cost, hand rig, soft soils to 30 m	Sample disturbed; unsuitable for gravel/boulders; slow
Percussion	Penetrates boulders & hard layers	Slow, disturbs soil below bit, no good sample
Rotary	Fast, deep, hard formations, core	Costly, heavy equipment, mud disposal

5.5 Stabilising the borehole

- *Casing:* steel pipe driven or placed to prevent caving; withdrawn after finishing (or left for piezometer).
- *Drilling mud (bentonite slurry):* forms a thin mud cake on the wall, hydrostatic pressure holds the wall open - no casing required in many soils; density ~ 1.05 - 1.2 t/m3.
- *Keep water level in casing higher than ground water table* to avoid *quick condition / piping* and loosening of sand at the bottom (very important before SPT).
- *Backfill* all exploratory holes with cement-bentonite grout / cuttings after use to protect ground water & safety.

5.6 Field safety and good practice

- Locate underground utilities before boring (power cables, gas, water lines).
- Record *rig down-time*, drilling rate, losses of drilling fluid (sign of cavities or fractured rock).

- Mark borehole coordinates with GPS/ total station and reduce level (R.L.) to a benchmark.

6. Summary table

Soil	Boring	Sampling
Soft clay	Auger / wash boring	Thin-wall tube / piston sampler
Stiff clay	Rotary / auger	Denison, Pitcher, block sample
Sand (dry)	Auger / hollow stem	Split-spoon (SPT) - disturbed
Sand below W.T.	Wash / rotary with mud	Split-spoon, piston (specialised), freezing
Gravel, boulders	Percussion	Bulk sample, scraper bucket
Rock	Rotary core drilling	Diamond core barrel (NX)

7. Important questions

- Describe various methods of boring. Explain wash boring with a neat sketch.
- Write short notes on types of augers and their uses.
- What is an undisturbed sample? Explain how disturbance is reduced in a sampler (area ratio, clearances).
- Explain the procedure of handling, storing and transporting samples.
- A thin-wall sampler has De = 76 mm, Di = 72 mm. Compute area ratio and comment. (Ar = 11.4 %)
- Differentiate split-spoon and Shelby tube sampler.

UNIT - III : BOREHOLE LOGGING

Logging methods | Groundwater observations | W.T. fluctuation | Soil profiles | Exploration report

> Borehole log = the written + graphical record of what the drill finds at every depth. It is the most important output of exploration - all design decisions and later contract claims are based on it.

1. Logging of boreholes

Logging means systematic recording of soil/rock encountered, the drilling operation, samples taken, tests performed and ground water observations. It is carried out by a trained geotechnical engineer / geologist at the rig - not from memory at the office.

1.1 What is recorded in a log?

Group	Items recorded
Header	Project, client, location, borehole number, co-ordinates, R.L. of ground, date, type of rig, diameter, driller, logged by
Strata	Depth of top & bottom of each layer, thickness, soil description (colour, consistency/density, grain size, plasticity, moisture, smell, organic content, structure), graphic symbol, IS/USCS group symbol
Sampling & tests	Depth and type of sample (D = disturbed, U = undisturbed, S = SPT, C = core), SPT N-value (15 cm blows each), vane, PMT, etc.
Drilling record	Method, casing depth, drilling fluid loss, rate of penetration, hard layers, boulders, cavities, drilling difficulties, time taken
Water	Depth of water first met, rest water level after 24 h, casing water level, artesian flow/seepage
Rock core	Core run, TCR, RCR, RQD, weathering grade, discontinuities, rock type

1.2 Soil description - the standard order

Write in this order (IS:1498 / BS 5930): Consistency or density -> colour -> secondary constituents -> PRIMARY SOIL -> particle shape / structure -> origin/other remarks. Example: "Stiff, brown, slightly sandy, silty CLAY with calcareous nodules (CI), residual."

Coarse soil	N (SPT)	Density term
Sand/gravel	0 - 4	Very loose
	4 - 10	Loose
	10 - 30	Medium dense
	30 - 50	Dense
	> 50	Very dense

Fine soil	N (SPT)	qu (kPa)	Consistency
Clay/silt	< 2	< 25	Very soft
	2 - 4	25 - 50	Soft
	4 - 8	50 - 100	Medium
	8 - 15	100 - 200	Stiff
	15 - 30	200 - 400	Very stiff
	> 30	> 400	Hard

Stages of a subsurface investigation programme

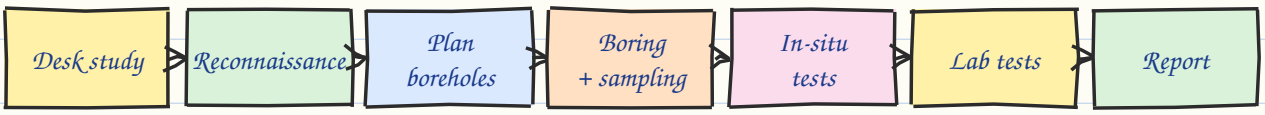


Fig 3.0 Programme of investigation

1.5 Tabular log (field sheet example)

Depth (m)	Description	Sample	SPT N (15-15-15 cm)	W.T.
0.0 - 0.5	Filled soil, brown, loose, with brick bits	D1 @ 0.5	-	-
0.5 - 2.0	Stiff brown silty clay (CI)	U1 @ 1.5	3-4-5 (N = 9)	-

Depth (m)	Description	Sample	SPT N (15-15-15 cm)	W.T.
2.0 - 4.5	Medium dense yellowish brown fine to medium sand (SP)	S2 @ 3.0	8-10-12 (N = 22)	3.2 m (24 h)
4.5 - 7.0	Dense gravelly sand with cobbles, GW	S3 @ 6.0	18-25-30 (N = 55)	-
7.0 - 8.0	Highly weathered granite (W4)	C1 (core run 1.0 m)	Refusal N > 100	-
8.0 - 11.0	Moderately weathered granite, TCR 85 %, RQD 55 %	C2	-	-

Note on SPT recording: The first 15 cm is the seating drive; N = blows for the next 30 cm (second + third 15 cm). If N > 50 for less than 30 cm penetration, record "Refusal, N > 50 for x cm" and stop.

1.6 Common mistakes and good logging practice

- Describing the soil only from the washed cuttings - always see the actual sample.
- Not giving the depth reference (ground level, casing top, platform).
- Writing "clay" for a silt or a clayey sand - do the thread test, dilatancy, dry strength, and wet-feel.
- Missing thin sand/silt partings in clay - drainage conditions depend on them.
- Giving N values without noting the hammer type, casing, and water level.
- Skipping the time and date of W.T. observation.

1.3 Graphical log (typical)

Depth (m)	Symbol	Description of strata	SPT N	Sample type / No.	W.T.
0.9		Brown silty clay, soft			
2.2		Grey clay, medium	7	SPT / D3	
3.6		Fine sand, loose	12	SPT / D6	
5.0		Medium sand, dense	31	SPT / D9	

Project: _____
BH No: 3
RL: 100.00 m
WT: 1.0 m
Date: _____

Fig 3.1 Format of a borehole log

1.4 Conventional hatch symbols

Soil / rock	Symbol (draw like this)
Gravel	Open circles (o o o) irregular
Sand	Fine dots (. . . .)
Silt	Short broken horizontal lines (- - -)
Clay	Continuous horizontal / diagonal lines
Peat / organic	Wavy lines with small vertical ticks
Rock	Bricks / zig-zag (V V V)
Filling	Cross-hatched triangle

2. Logging methods

Method	Description	Remarks
Visual logging of cuttings / samples	Strata identified from auger cuttings, washed material, SPT split-spoon sample, tube sample	Standard method; accuracy depends on skill
Drilling-time / penetration rate log	Rate of advance or drilling pressure plotted with depth (change in rate = change of strata)	Used in percussion/rotary where samples are scarce
Core logging	Rock cores laid out in core boxes: TCR, SCR, RQD, fractures, weathering	For rock and hard soil
Continuous sampling	Continuous tube sampling (e.g. Delft, Sonic) records every layer	Expensive, used for critical projects

Method	Description	Remarks
Geophysical borehole logging	Probes lowered in hole: natural gamma, resistivity, SP, density, caliper, sonic, televiewer	Good for correlation and for holes where core was lost
In-situ test logs	Cone resistance and friction ratio (CPT), N vs depth (SPT)	Continuous profile of soil stiffness

Rock core terms: TCR = Total core recovery = (length of core recovered / length of run) x 100. SCR = Solid core recovery (pieces >= full dia. of core). RQD = sum of core pieces longer than 10 cm / length of run x 100. RQD: 0-25 very poor, 25-50 poor, 50-75 fair, 75-90 good, 90-100 excellent.

Example: In a 1.5 m core run the pieces > 10 cm are 20, 15, 30, 12, 18 cm (sum = 95 cm). Recovery 1.2 m. TCR = 1.2/1.5 = 80 %. RQD = 95/150 x 100 = 63 % (fair rock).

2.1 Logging of rock cores (IS:4464)

Weathering grade	Term	Description
W1	Fresh	No visible sign of weathering
W2	Slightly weathered	Discoloration along joints; rock strong
W3	Moderately weathered	Less than half rock decomposed; core stones present
W4	Highly weathered	More than half decomposed to soil; rock fragments
W5	Completely weathered	All rock changed to soil, original structure visible
W6	Residual soil	Original structure destroyed

Joint spacing	Term
> 2 m	Very wide
0.6 - 2 m	Wide
0.2 - 0.6 m	Moderate
60 - 200 mm	Close

Joint spacing	Term
< 60 mm	Very close

Also record: rock name (from mineral/grain), colour, strength (field test: hammer blow - very weak: crumbles; weak: scratched with knife; strong: needs several blows), fracture frequency per metre, dip of joints (degrees from horizontal), infilling of joints (clay, calcite), and water loss.

2.2 Geophysical (wire-line) logging in boreholes

Log	Measures	Used to find
Natural gamma	Radioactivity of strata	Clay / shale (high) vs sand (low) - lithology
Electrical resistivity	Resistance of formation + fluid	Saturation, water quality, sand/clay
Spontaneous potential (SP)	Natural electric potential	Permeable zones
Caliper	Hole diameter	Caving zones, fractures
Sonic (velocity)	P-wave velocity	Rock quality, porosity, fractures
Density/neutron	Bulk density, moisture	Porosity of strata
Televiewer / camera	Image of borehole wall	Joints, dip & strike, cavities

3. Ground water observations

Water table (W.T.) or phreatic surface is the level at which the pore pressure is atmospheric. Its position changes with season, rainfall, pumping and tides, so it must be observed carefully and repeatedly during exploration.

3.1 How to observe the W.T. in a borehole

- Record the depth at which water is first encountered (moist -> wet sample, drilling fluid loss or gain) during advance.
- Stop drilling for the day and note the water level in the hole at the end of shift and again next morning before starting (after 24 h) - this is the "rest" or stabilised W.T. For clay the time is longer (even 1 - 2 weeks).

- In sandy soils the water in a hole comes into equilibrium within minutes to hours; in clays after days.
- Remove casing or bail the hole and watch the recovery; use a dip meter (water level indicator) or a tape with a probe sounding.
- For long term monitoring install an observation well / standpipe or a piezometer (Unit V).
- Record any artesian pressure, perched water table, and whether the water is flowing from fissures or from sand lenses.

3.2 Types of ground water

Type	Meaning
Unconfined (phreatic) aquifer	W.T. free; rises & falls with recharge. Pore pressure = $\gamma_w \times$ depth below W.T.
Perched water table	A small body of water held above the main W.T. by a thin impermeable lens
Confined / artesian aquifer	Water under pressure between two impervious layers; piezometric level may be above ground
Capillary water	Held above W.T. by surface tension: $h_c = 0.3/d$ (cm), clays can rise several metres, sands 0.1 - 1 m

3.3 Water table fluctuations and their effects

W.T. changes due to: rainfall / monsoon recharge, evaporation, pumping of wells, tides, reservoir level, irrigation canals, leaking pipes, temperature and barometric changes. Annual variation of 1 - 3 m is common in India; foundations are always designed for the highest likely W.T.

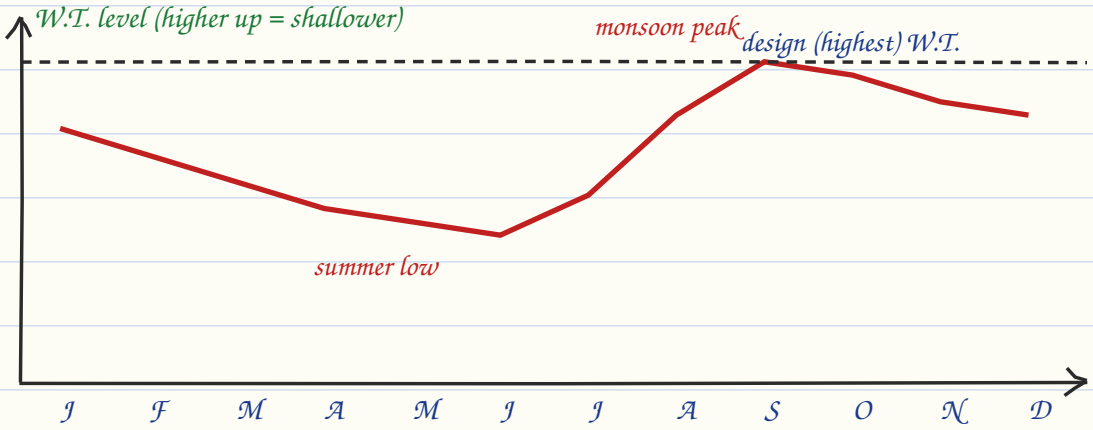
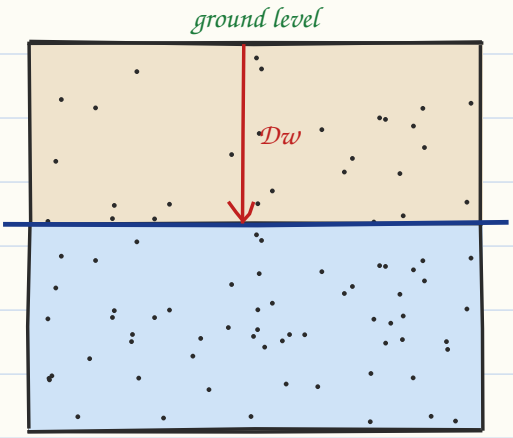


Fig 3.2a Typical seasonal fluctuation of ground water table



Effect of W.T. rising to ground:

- effective stress $s' = s - u$ decreases
- bearing capacity drops by upto 50 %
- settlement and heave increases
- buoyancy / uplift on basements
- seepage, quick sand, piping
- dewatering cost rises

Fig 3.2 Effect of a rise in W.T.

Effect	Explanation
Effective stress	Rise of W.T. reduces effective stress: $s' = s - u$. Strength and bearing capacity fall.
Bearing capacity	Terms $\gamma D f N_q$ and $0.5 \gamma B N_\gamma$ use submerged unit weight $\gamma' = \gamma_{sat} - \gamma_w$ (about half); capacity reduces up to 50 %. Correction factor $R_{w1}, R_{w2} = 0.5 + 0.5\, D_w/D_f$.
Settlement	Consolidation settlement of clays increases when W.T. falls (effective stress increases) - e.g. due to pumping: land subsidence
Swelling / heave	Wetting of expansive soils (black cotton) with rising W.T.
Uplift / buoyancy	Basements, tanks, and pile caps floating; need tension piles/ weight
Excavation problems	Seepage into excavations, quick sand, piping, loosening - need well-point dewatering
Corrosion & chemical attack	Sulphates/chlorides in ground water attack concrete & steel

Effect	Explanation
Liquefaction potential	High W.T. in loose sand is a main factor in earthquake liquefaction

SPT correction for water: in saturated fine/silty dense sands (N > 15) apply N_corrected = 15 + 0.5 (N_observed - 15) (Terzaghi & Peck).

3.4 Effect of W.T. on stress - worked example

Problem. Soil: gdry = 16 kN/m3 above W.T., gsat = 19 kN/m3. Find vertical effective stress at 6 m depth when (a) W.T. at 2 m, (b) W.T. rises to ground level. (gw = 9.81)

(a) total s = 16 x 2 + 19 x 4 = 108 kPa; u = 9.81 x 4 = 39.2 kPa; s' = 68.8 kPa

(b) s = 19 x 6 = 114 kPa; u = 9.81 x 6 = 58.9 kPa; s' = 55.1 kPa

Rise of W.T. reduces effective stress by 20 % -> strength and bearing capacity fall.

Capillary rise example. Effective grain size D10 = 0.02 mm (silty sand): hc = C/(e D10) with C = 0.3 cm2, e = 0.6 -> hc = 0.3/(0.6 x 0.002) = 250 cm = 2.5 m. Soil in this zone is saturated but with negative pore pressure (suction) -> apparent cohesion.

3.5 Quality of ground water (tests on 1 L sample)

Parameter	Concern	Limit (IS:456 for concrete)
pH	Acidic water attacks concrete	>= 6
Sulphates (as SO4)	Sulphate attack on cement	< 400 mg/l (mix) ; ground: 0.2 % soluble in soil
Chlorides	Corrosion of reinforcement	< 500 mg/l (RCC); 2000 (PCC)
Organic matter, TDS	Setting time, strength	TDS < 2000 mg/l

3.6 Controlling W.T. during construction

- Sump pumping / open pumping - for small seepage in clays & coarse soils.
- Well points - for sands (1 - 6 m), suction lift; deep wells for deep drawdown.
- Cut-off walls (sheet piles, diaphragm walls, grouting) - reduce flow into excavation.
- Freezing or electro-osmosis for silts and very fine soils.
- Always lower W.T. at least 0.5 - 1 m below the excavation level and avoid loss of fines.

4. Preparation of soil profiles

A soil profile (subsoil section) is a vertical section drawn through two or more boreholes showing the sequence of strata, thickness, W.T. and test results. It gives a picture of the ground for design and construction.

4.1 Steps in preparing a profile

- Plot boreholes at horizontal scale (1:200 to 1:500) along the alignment/section line; vertical scale larger (1:50 to 1:100) = exaggerated.
- Reduce depths to R.L. using the ground levels.
- Mark the strata boundary at each borehole and join corresponding layers - using straight/dashed lines. Use dashed lines when the boundary is inferred (uncertain).
- Show pinching out, lenses and faults carefully; do not extend a boundary too far from the boreholes.
- Add graphical symbols, group symbols, N values, W.T. (with date), sample positions, foundation level.

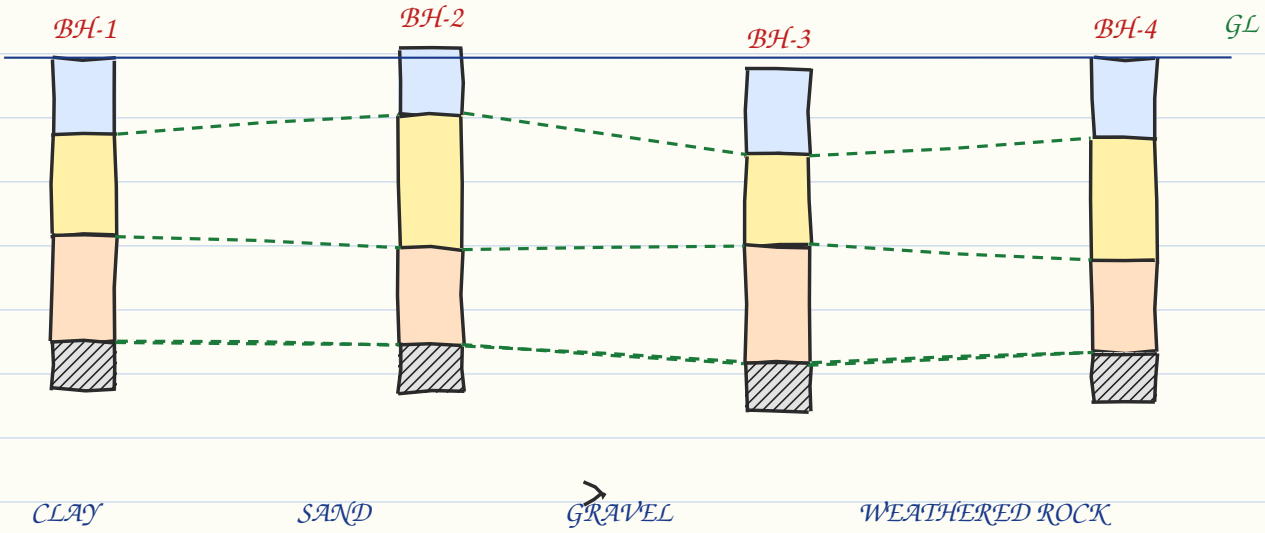


Fig 3.3 Subsoil section through four boreholes

5. Exploration (site investigation) report

The final document of an investigation, containing factual data and the interpretation, with recommendations. Details are in Unit V. Contents in short:

- Introduction: project, location, scope, dates, personnel, investigation method.
- Site description & geology: topography, drainage, regional geology, previous use.
- Field work: boring methods, depth, sampling, tests carried out, borehole logs (annexure).
- Ground water: depth, fluctuation, quality.
- Laboratory tests and results: tabulated.
- Analysis & recommendations: type and depth of foundation, safe bearing capacity / allowable pressure, settlement, special precautions (swelling soil, fill, dewatering).
- Drawings: site plan with borehole location, subsoil sections, graphs.

5.1 Specimen contents of an exploration report

Chapter	Heading
1	Introduction (client, project, objectives, standards followed)
2	Site and geology (location map, topography, drainage, climate, seismic zone)
3	Field investigations (borehole table, methods, in-situ tests, ground water)

Chapter	Heading
4	Laboratory test results (summary sheets, grain size curves, consolidation, shear)
5	Discussion of subsoil conditions (idealised soil profile, soil parameters)
6	Foundation recommendations (type, depth, SBC, settlement)
7	Construction considerations (dewatering, excavation support, special problems)
Annex	Borehole logs, laboratory data sheets, drawings, photographs

Typical recommendation (sample): "Based on the investigation, the site consists of 1.5 m filled soil underlain by stiff silty clay (CI) up to 6 m resting on weathered rock. W.T. was not met up to 12 m (April 2025). Isolated footings at 2.0 m depth below existing G.L. in the stiff clay may be designed for a net safe bearing capacity of 180 kN/m2, limiting settlement to 40 mm."

Do & do not

- Do record actual depths, never "typical" depth; note any change in drilling method.
- Do photograph cores and samples; label boxes.
- Don't leave out disturbed-layers (made-up ground) or boulders.
- Don't mix up borehole numbers or levels - an error can mislead the whole design.

6. Important questions

- Explain how a borehole is logged. Draw a typical borehole log.
- What are the observations to be made regarding ground water in boreholes?
- Explain the effects of water table fluctuations on foundation.
- How is a soil profile prepared from borehole data?
- Write the contents of a soil investigation report.
- Define TCR, SCR and RQD. Compute RQD for a given core run.

UNIT - IV : FIELD TESTING OF SOILS

SPT | PLT | PMT | DMT | VST | CPT | BPT | Geophysical methods

Why field tests? Undisturbed samples are expensive and hard to get in sands, gravels and soft clays. **In-situ tests** measure soil response directly in its natural state, give continuous profiles and let us estimate strength, stiffness and bearing capacity quickly.

1. Classification of field tests

Type	Tests	Gives
Identification	Visual / manual tests	Soil name, consistency
Penetration	SPT, DCPT, SCPT, CPT, BPT	Strength, density (N, qc)
Loading	Plate load test (PLT), pile load test	Bearing capacity, settlement
Expansion / shear	Pressuremeter (PMT), flat dilatometer (DMT), vane (VST)	Modulus, strength, stress state
Geophysical	Seismic refraction, resistivity, cross-hole	Strata boundaries, wave velocity, G
Permeability	Pumping test, Lefranc, packer (Lugeon)	k, groundwater flow

2. Visual identification tests (field identification)

Done on the spot by looking at the sample and doing simple hand tests. Needed for the boring log. (IS:1498 / IS:2720).

Test	Procedure	Result
Dry strength	Crush dried 12-mm pat between fingers	None (silt/sand); low; medium; high (CH clay)
Dilatancy (shaking) test	Wet pat of soil in palm, shake & squeeze	Rapid water appears = silt/fine sand; slow/none = clay

Test	Procedure	Result
Toughness (thread) test	Roll a thread to 3 mm, fold, re-roll near plastic limit	Tough thread = high plasticity clay; weak = silt
Dispersion test	Drop soil in water cylinder, observe settling	Sand settles in 30 s, silt in 30 min, clay stays cloudy
Shine / smear test	Cut with knife	Shiny surface -> clay
Feel & odour	Rub between fingers; smell	Gritty = sand; smooth = silt; sticky = clay; organic odour = peat
Sedimentation, bite test	Gritty on teeth = silt/sand	Quick sizing

Consistency tests for clay: thumb pressed easily - soft; moulded by strong pressure - medium; thumb dents with difficulty - stiff; thumbnail can mark - very stiff; hard - nail cannot mark.

3. Standard Penetration Test (SPT) - IS:2131-1981

The most common test in India. A split-spoon sampler is driven into the soil at the bottom of a borehole by a 63.5 kg hammer falling 750 mm. The number of blows for the last 300 mm of the 450 mm drive is the N value.

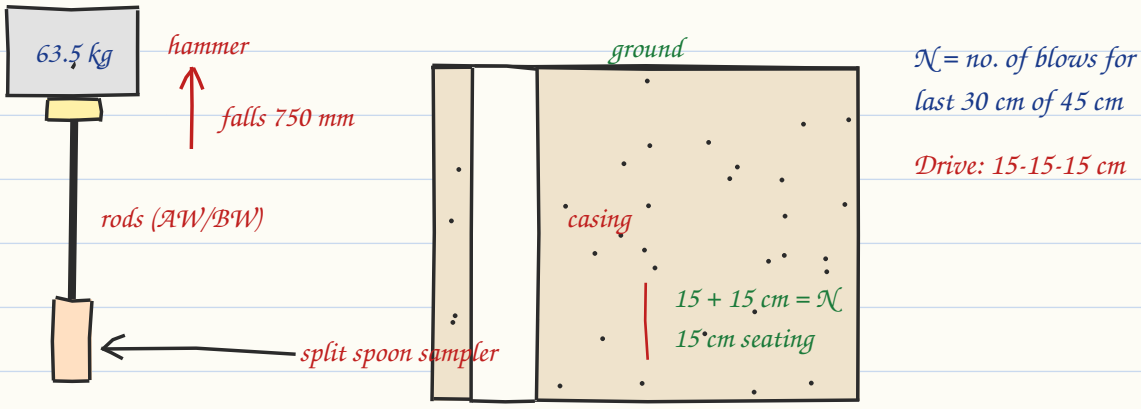


Fig 4.1 Standard penetration test

3.1 Procedure

- Advance borehole to test depth (usually every 1.5 m or change of strata); clean the hole; keep water in hole higher than W.T.
- Lower the split spoon (50.8 OD x 35 ID) on rods; seating drive of 150 mm - record blows.
- Drive next two 150 mm increments - add blows = N. Stop if 50 blows in any 150 mm increment -> refusal (note penetration).
- Pull out the spoon, open it, describe & preserve the sample in jar (SPT sample = disturbed representative).
- Record N for each depth in the log.

3.2 Corrections to N (Peck, Hanson & Thornburn)

Correction	Expression	Remarks
Overburden (CN)	CN = 0.77 log10 (2000/s0') [s0' in kPa], CN <= 2	For sands; N1 = CN x N
Energy / hammer	N60 = N x (ER/60)	Donut hammer ER ~ 45 %, safety 60 %, automatic 80 %
Rod length	Cr < 1 for short rod (< 10 m): 0.75 (3-4 m), 0.85 (4-6 m), 0.95 (6-10 m)	Energy loss in rods
Borehole diameter	Cb = 1.0 (65-115 mm), 1.05 (150 mm), 1.15 (200 mm)	
Sampler liner	Cs = 1.0 (with liner), 1.2 (without)	
Dilatancy (water)	N = 15 + 0.5 (N - 15) for fine/silty sands below W.T. with N > 15	Terzaghi-Peck

3.3 Correlations (Terzaghi & Peck)

Sand N	Density	phi (deg)	Clay N	Consistency	qu (kPa)
0 - 4	Very loose	< 29	< 2	Very soft	< 25
4 - 10	Loose	29-30	2 - 4	Soft	25 - 50
10 - 30	Medium	30-36	4 - 8	Medium	50 - 100
30 - 50	Dense	36-41	8 - 15	Stiff	100 - 200

Sand N	Density	phi (deg)	Clay N	Consistency	qu (kPa)
> 50	Very dense	> 41	15 - 30	Very stiff	200 - 400
			> 30	Hard	> 400

- Clay: cu = qu/2 ~ 6.25 N kPa (Terzaghi) or cu = 6 N roughly.
- Allowable bearing pressure of sand from N via Meyerhof / IS:8009 charts (settlement 25 mm).
- Used also for liquefaction screening, pile capacity (end bearing 40 N60 x Db/B kPa).

Limitations: crude, disturbed sample, depends on hammer energy, operator and boring method; unreliable in gravels (oversize), soft clays (low N), and fine sand below W.T. without care.

4. Dynamic cone penetration test (DCPT) - IS:4968

A cone (50 mm dia, 60 deg) is driven by 65 kg hammer (falling 750 mm) without boring. Blows per 300 mm (Ncd) are recorded. Bentonite slurry can be used for friction reduction in the 100-mm type with casing. N value ~ 1.5 Ncd for fine sand; very cheap for quick checking of the foundation of low-rise structures.

5. Plate Load Test (PLT) - IS:1888-1982

Direct test of bearing capacity and settlement: a rigid steel plate placed at foundation level is loaded in increments and settlement measured. Reaction by kentledge (dead load) or anchors / loaded truck.

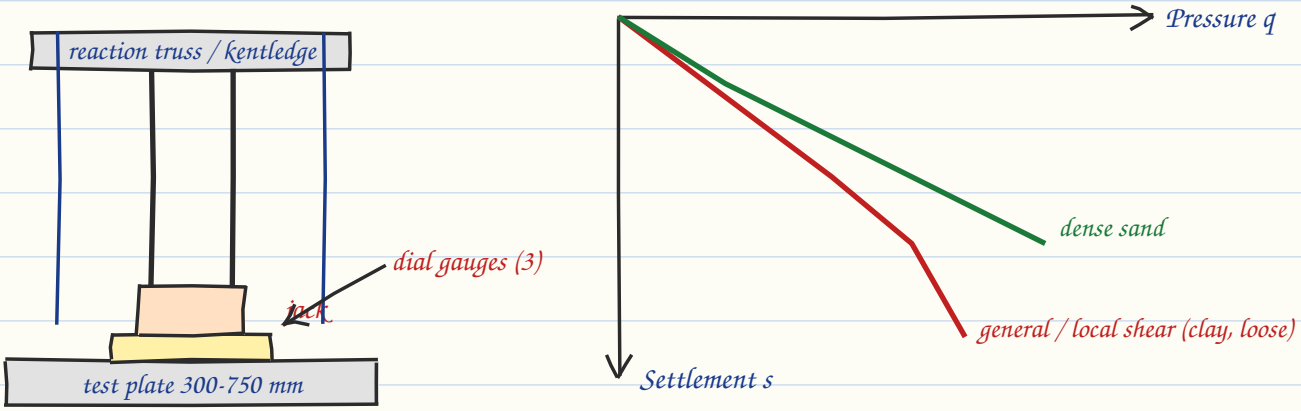


Fig 4.2 Plate load test and typical load-settlement curves

5.1 Procedure

- Excavate a pit of width not less than 5 x plate size, to proposed foundation level; level & clean bottom (fine sand bed).
- Plates: square 300, 450, 600 mm or circular 300-750 mm dia, thickness 25 mm.
- Apply load in increments of 1/5 of the estimated safe load (or 1/10 of ultimate); each stage until settlement rate < 0.02 mm/min (or 1 h) ; settlement by 2 - 4 dial gauges (0.01 mm).
- Continue till failure or 2.5 x design load, or settlement 25 mm (sand) / 10 % of plate width.
- Plot load-settlement curve (log-log or arithmetic).

5.2 Interpretation (size effect)

Soil	Ultimate bearing capacity	Settlement of foundation (Sf) from plate (Sp)
Clay	qf = qp (independent of size)	Sf = Sp x (Bf/Bp)
Sand	qf = qp x (Bf/Bp)	Sf = Sp x [Bf (Bp + 0.3) / (Bp (Bf + 0.3))]^2 (B in m)
c-phi soil	Use equations from c and phi of the soil	-

Example: 0.3 m plate in sand fails at 280 kN/m2; footing width 2 m. qf = 280 x 2/0.3 = 1867 kN/m2 (too optimistic; take lower) ; settlement for plate 12 mm at 100 kN/m2 -> foundation: 12 x [2 x 0.6/(0.3 x 2.3)]^2 = 12 x (1.739)^2 = 36.3 mm.

- Limitations: reflects only soil within 1.5 - 2 plate widths below plate; scale effect, for consolidation of deep clays use the consolidation test; time-consuming/expensive.
- Plate bearing test for pavements uses 300-750 mm plate to find modulus of subgrade reaction k = q/delta (kN/m3).

6. Pressuremeter test (PMT) - Menard

A cylindrical probe (flexible rubber membrane) is lowered in a borehole and expanded radially by pressurised water/gas; volume change vs pressure is measured. Gives in-situ stress-strain behaviour.

Parameter	Meaning / formula
p0	At-rest horizontal pressure (start of linear part)
pf	Creep pressure (end of pseudo-elastic part)
pL	Limit pressure (volume doubles) - used for bearing capacity: qu = k (pL - p0) + sv0
EM	Pressuremeter modulus EM = 2 (1 + n)(V0 + Vm)(Dp/DV)
Cu (clay)	cu = (pL - p0)/Np, Np ~ 5.5

- Advantages: modulus & strength in all soils and weak rock; good for deep foundations and settlement.
- Types: Menard (pre-bored), Self-boring pressuremeter (SBPM) - minimal disturbance, High-pressure dilatometer for rock.

7. Flat Dilatometer Test (DMT) - Marchetti

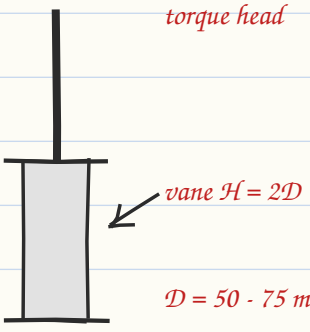
A steel blade (14 mm thick, 95 mm wide) with a thin circular steel membrane (60 mm) on one face is pushed (or driven) into soil at 20 mm/s. At each 200 mm depth, gas pressure inflates the membrane. Two readings: A (lift-off pressure, membrane begins to move) and B (pressure at 1.1 mm expansion).

Index	Formula	Gives
Material index ID	(p1 - p0)/(p0 - u0)	Soil type: <0.6 clay, 0.6-1.8 silt, >1.8 sand
Horizontal stress index KD	(p0 - u0)/sv0'	OCR, K0 (K0 ~ (KD/1.5)^0.47 - 0.6)
Dilatometer modulus ED	34.7 (p1 - p0)	Constrained modulus M = RM ED
Undrained strength cu	0.22 sv0' (0.5 KD)^1.25	Clays

p0 = 1.05 (A - ZM + DA) - 0.05 (B - ZM - DB) ; p1 = B - ZM - DB. Fast (about 1 hour / 20 m), repeatable, cheap; widely used for settlement and liquefaction.

8. Vane Shear Test (VST) - IS:4434

The best in-situ test for undrained shear strength (cu) of soft, sensitive clays and silts. A four-bladed vane (H/D = 2) is pushed into the soil and torque applied slowly; torque at failure T gives the strength of the cylindrical surface.



Su = T / (pi D2 (H/2 + D/6))

T = max torque at failure (N.m)

Rate: 6 - 12 deg per min

Remoulded: rotate 10 turns, retest

Sensitivity St = Su / Sur

Only for soft & medium clays (Su < 100 kPa)

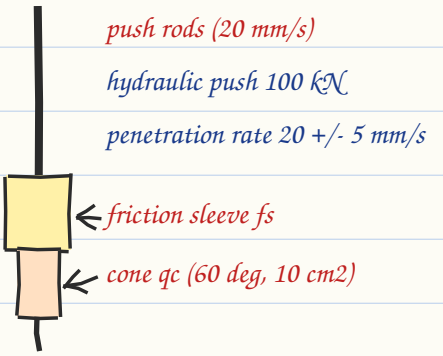
Fig 4.3 Vane shear test

Formula: T = cu (pi D2 H/2 + pi D3/6) -> cu = T / [pi D2 (H/2 + D/6)]. For H = 2D: cu = 6T/(7 pi D3). Example: D = 75 mm, H = 150 mm, T = 62 N.m -> cu = 62 / [pi x 0.075^2 x (0.075 + 0.0125)] = 62 / (0.017671 x 0.0875) = 40 kPa (soft clay).

- Bjerrum correction (clays with Ip high): cu(design) = m x cu(vane), m = 1.7 - 0.54 log Ip (decreasing with plasticity).
- Steps: push vane 0.5 m below borehole bottom; rotate 6 - 12 deg/min; record T vs angle; peak = undisturbed; rotate 10 turns -> remoulded; test again -> St.
- Not suited to stiff clays, sands, gravels or fissured clays.

9. Cone Penetration Test (CPT / Dutch cone) - IS:4968 Pt3

A standard 60-degree cone of 10 cm2 base area is hydraulically pushed at constant 20 mm/s into the soil. Measures cone resistance qc and sleeve friction fs (mechanical or electric cone). With a pore-pressure transducer = piezocone (CPTu).



Outputs:

qc cone resistance (MPa)
fs sleeve friction (kPa)
Rf = fs/qc x 100 (friction ratio)
u2 pore pressure (CPTu)
Sand: high qc, low Rf (< 1 %)
Clay: low qc, high Rf (3 - 6 %)

Fig 4.4 Cone penetration test

Soil	qc (MPa)	Rf (%)	Remarks
Soft clay	0.2 - 1	3 - 6	Higher Rf
Stiff clay	1 - 4	2 - 4	
Loose sand	2 - 5	< 1	
Dense sand	10 - 30	0.5 - 1.5	
Gravel	> 30	< 1	Refusal common

- Correlations: N = qc/ (0.4 MPa to 0.6 MPa) (qc in kg/cm2 = 4 N for sand, 2 N for clay-sand); cu = (qc - sv0)/Nk, Nk = 15 - 20; Es = a qc (a = 2 - 4 sand).
- Bearing capacity of piles (end resistance qb ~ qc), liquefaction analysis, soil classification (Robertson chart, Ic).
- Advantages: continuous profile, fast, repeatable, no sample/hole, cheap. Limitations: no samples; refusal in gravel; need calibration with boreholes.

10. Becker Penetration Test (BPT)

Used in gravelly soils/ cobbles where SPT fails (oversized particles). A double-walled 168 mm closed-end casing (Becker drill) is driven continuously by a diesel pile hammer (rate 55-60 blows/min); BPT = no. of blows for each 300 mm advance. BPT values are converted to equivalent SPT N by Harder & Seed correlations using bounce-chamber pressure to correct for hammer energy:

- Corrected N-equivalent = f(Nb, bounce chamber pressure) ; for liquefaction assessment of gravelly sands/dams.
- Pros: penetrates gravels, up to 150 m; Cons: complicated corrections, no sample in closed type (open-ended Becker allows reverse circulation sampling).

11. Geophysical methods of exploration

Indirect, non-destructive methods using physical properties (velocity, resistivity) of ground. Cover large areas rapidly at low cost; must be calibrated by boreholes.

11.1 Seismic refraction method

A shock (sledgehammer / explosive) generates P-waves; geophones record first arrival times. Waves refracted at the interface with a faster layer (V2 > V1) return to surface and arrive earlier than direct wave beyond the critical distance (Xc).

Critical angle ic = sin-1 (V1/V2)

Depth z = (Xc/2) sqrt((V2-V1)/(V2+V1))

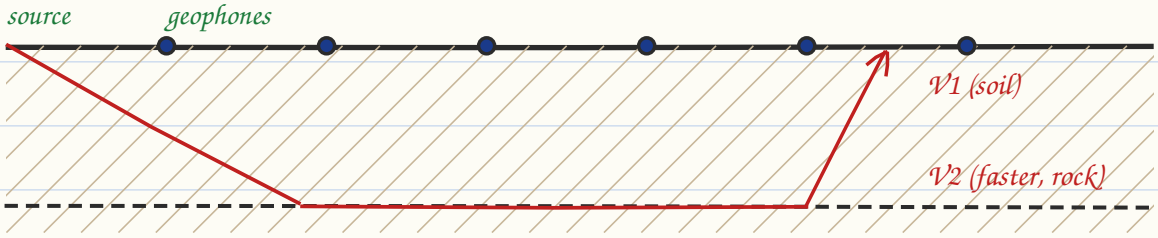


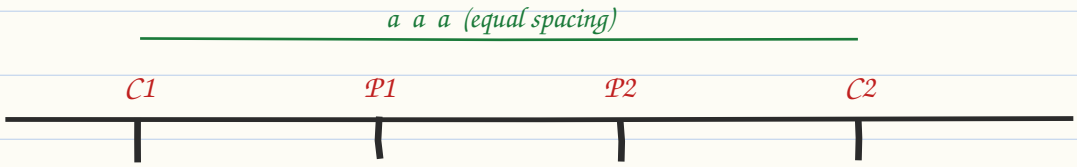
Fig 4.5 Seismic refraction

- Plot time-distance graph; two straight segments; slope = 1/V. Intercept time Ti gives depth: z = Ti V1 V2 / (2 sqrt(V2^2 - V1^2)).
- Typical P-wave velocities (m/s): dry sand 200-1000; wet sand 1000-2000; clay 1000-2500; water 1500; weathered rock 2000-4000; hard rock 4000-6000.
- Limitations: V must increase with depth (hidden layer / velocity reversal not detected).

11.2 Electrical resistivity method

Current is passed through ground by two current electrodes; the potential drop across two potential electrodes gives apparent resistivity. Resistivity depends on moisture, salt content and porosity: dry rock very high, clay & wet soil low.

Current I through C1-C2, potential V across P1-P2:



rho = 2 pi a (V/I) (a = electrode spacing)

Schlumberger: rho = pi (L2 - l2)/(2 l) x V/I, L = AB/2, l = MN/2

Fig 4.6 Wenner and Schlumberger arrays

Material	Resistivity (ohm-m)
Clay, saturated	1 - 100
Wet sand / gravel	100 - 1000
Dry sand	1000 - 5000
Sea water	0.2
Granite, massive rock	10^3 - 10^6

- *Sounding (VES):* electrodes expanded about a centre -> depth increases; *profiling:* electrodes shifted along line at fixed spacing - for lateral variation, buried channel.
- Uses: depth of W.T., bed rock, clay lenses, corrosion & earthing design, location of cavities; limited by electrical interference.

11.3 Cross-hole test

Two or three boreholes at 3 - 5 m spacing; a seismic source (hammer/ SH-wave source) in one hole, receivers (geophones) at same depth in other holes - measures shear (S) wave velocity Vs directly at each depth. (ASTM D4428). Holes are cased with PVC, grouted, inclinometer survey to know exact spacing.

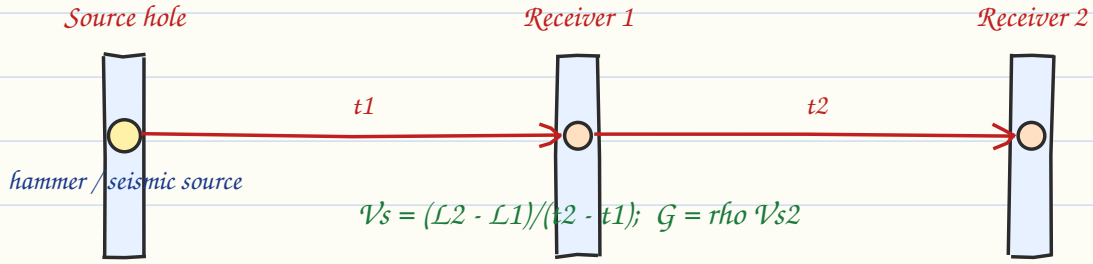


Fig 4.7 Cross-hole test

Calculations: Vs = (L2 - L1)/(t2 - t1) ; Gmax = r Vs ^ 2 ; Emax = 2 (1 + n) Gmax. Used for dynamic analysis, machine foundations, earthquake site response. Down-hole and up-hole tests are cheaper variants using one borehole.

12. Analysis of test results - summary

Test	Result used for	Typical design use
SPT	N60, phi, cu, density	Bearing capacity, pile design, liquefaction
PLT	q-s curve, k	Safe bearing pressure of shallow foundation
PMT	pL, EM	Foundation capacity, deformation
DMT	ID, KD, ED	OCR, K0, settlement
VST	cu, sensitivity	Soft clay embankments, stability

Test	Result used for	Typical design use
CPT	qc, fs, u	Soil profile, piles, liquefaction
BPT	Becker N	Gravel / liquefaction
Seismic refraction	V, depth of rock	Rippability, rock head
Resistivity	rho	W.T., rock, corrosion
Cross-hole	Vs, Gmax	Dynamic soil properties

13. Important questions

- Explain the SPT procedure and corrections applied to N values.
- Describe plate load test. How do you find the bearing capacity & settlement of footing from it?
- Write short notes on PMT and DMT.
- Explain the vane shear test and its limitations.
- What is CPT? List its advantages over SPT.
- Write about Becker penetration test.
- Explain seismic refraction and electrical resistivity methods for subsurface exploration.
- Describe cross-hole test and how shear wave velocity is determined.

UNIT - V : REPORT WRITING & FIELD INSTRUMENTATION

Soil exploration reports | Strain gauges | Piezometers | Pressure cells | Inclinometers | Proving ring | Load cells | Displacement gauges

Two parts of this unit: (A) the soil exploration report - how to identify, calculate and present the findings; (B) field instrumentation - devices used to monitor the performance of soil, rock and structures during and after construction.

PART A - SOIL EXPLORATION REPORT

1. Purpose and importance

The report is the final product of a soil investigation. It communicates the subsoil conditions to the designer and contractor, and recommends the type and depth of foundation and allowable bearing pressure. It is also a legal document - claims for "differing site conditions" refer to it. (IS:1892-1979 "Code of practice for subsurface investigation for foundations").

Qualities of a good report

- Clear, concise, systematic - arranged in logical order with an index.
- Factual data separated from interpretation.
- Sufficient number of boreholes with plan, logs, sections.
- Gives design parameters (c, phi, gb, Es, cv) with ranges, not single numbers.
- States assumptions and limitations, date of W.T. observation.
- Signed and dated by a responsible engineer.

2. Identification - what must be identified?

Item	Identification
Project	Name, client, type of structure, column loads, floor levels, area of site, proposed foundation level

Item	Identification
Site	Location map (with lat-long), topography, vegetation, adjacent structures, drainage, flooding, previous land use (fill, tank, quarry)
Geology	Geological map, rock/soil origin, faults, earthquake zone (IS:1893), special soils (black cotton, laterite, collapsible, peat)
Soil strata	Layer-wise thickness, type, group symbol, colour, consistency/density, special features
Ground water	Depth, date, fluctuation, flow direction, chemical analysis
Samples & tests	Locations, depth, quantity, standards (IS:2720 parts) followed

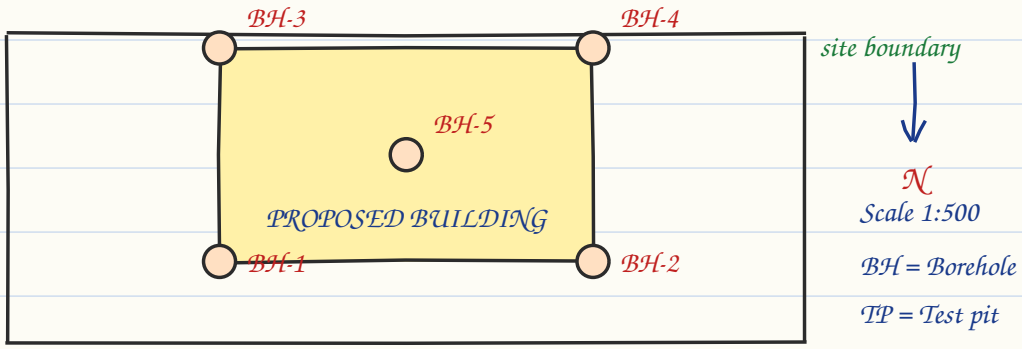


Fig 5.1 Borehole location plan

3. Contents and format of the report

Section	Contents
1. Title page & index	Project name, report no., date, consultant, list of contents/figures/tables
2. Summary	One-page: soil profile, W.T., recommended foundation, SBC, major risks
3. Introduction	Scope, terms of reference, structure details, standards used
4. Field work	Boring method, equipment, depth of each BH, sampling, SPT, in-situ tests
5. Lab work	Tests: water content, G, grain size, Atterberg, shear (UCS/ triaxial/ direct), consolidation, CBR, chemical
6. Subsoil description	General profile, individual layer description, W.T.
7. Analysis	Bearing capacity, settlement, pile capacity, slope stability etc.
8. Recommendations	Foundation type, depth, allowable pressure, precautions, dewatering, filling, retaining wall

Section	Contents
9. Annexures	Bore logs, test data sheets, graphs, photographs, drawings, borehole location plan, soil profiles

4. Calculations in the report

4.1 Index properties summary

Tabulate for each sample: depth, w, gd, G, Sr, e, wL, wP, Ip, % gravel/sand/silt/clay, IS classification; and strength: c, phi (or cu), compressibility: Cc, cv, e0.

4.2 Safe bearing capacity (Terzaghi / IS:6403)

Terzaghi (square footing, general shear): $q_u = 1.3\,c\,N_c + \gamma\,D_f\,N_q + 0.4\,\gamma\,B\,N_\gamma$; strip: $q_u = c\,N_c + \gamma\,D_f\,N_q + 0.5\,\gamma\,B\,N_\gamma$; circular: $1.3\,c\,N_c + \gamma\,D_f\,N_q + 0.3\,\gamma\,B\,N_\gamma$.

Net ultimate $q_{nu} = q_u - \gamma\,D_f$. Net safe $q_{ns} = q_{nu} / F$ (F = 2.5 to 3). Safe $q_s = q_{ns} + \gamma\,D_f$.

Local shear failure (soft/loose): use $c' = 2c/3$, $\tan\phi' = (2/3)\tan\phi$.

Example 1. Square footing B = 2 m at D_f = 1.5 m in clay (ϕ = 0): c = 40 kPa, γ = 18 kN/m3 ; N_c = 5.7 (Terzaghi, ϕ=0), N_q = 1, N_γ = 0.

$q_u = 1.3 \times 40 \times 5.7 + 18 \times 1.5 \times 1 = 296.4 + 27 =$ 323.4 kPa

$q_{nu} = 323.4 - 27 = 296.4$; $q_{ns} = 296.4 / 3 =$ 98.8 kPa ; safe pressure = 98.8 + 27 = 125.8 kPa.

Example 2 (sand from SPT, Meyerhof). Corrected N = 20, B = 2 m (> 1.2 m), D_f = 1.5 m, permissible settlement 25 mm.

$F_d = 1 + 0.33\,D_f/B = 1 + 0.33 \times 0.75 = 1.25$ (limit 1.33)

$q_a = 11.98\,N\,[(B + 0.3)/B]^2\,F_d = 11.98 \times 20 \times (2.3/2)^2 \times 1.25 = 11.98 \times 20 \times 1.3225 \times 1.25 =$ 396 kPa (check against shear failure and W.T. correction).

4.3 Settlement

- **Immediate:** $S_i = q B (1 - \nu^2) I_f / E_s$
- **Consolidation:** $S_c = C_c H/(1 + e_0) \times \log_{10}[(\sigma_0' + \Delta\sigma)/\sigma_0']$ for a normally consolidated clay layer thickness H.
- **Time:** $t = T_v H_{dr}^2 / c_v$ ($T_v = 0.848$ for 90 % consolidation).
- **Allowable:** isolated footing 40 mm (sand), 65 mm (clay); raft 75 mm; differential 1/500 to 1/300 (IS:1904).

> **Example 3.** 4 m thick NC clay: $e_0 = 0.9$, $C_c = 0.3$, $\sigma_0' = 80$ kPa at mid-depth, $\Delta\sigma = 40$ kPa. $S_c = 0.3 \times 4/(1.9) \times \log_{10}(120/80) = 0.632 \times 0.1761 = 0.1112$ m = 111 mm.

4.4 Foundation recommendation logic

Soil condition	Recommended foundation
Good bearing soil within 1.5 - 3 m	Isolated / strip footing
Weak soil near surface, firm below 5 - 6 m	Raft, pile or under-reamed pile
Black cotton / swelling soil	Under-reamed piles, sand cushion, belled piers, CNS layer
Very soft clay/ high W.T.	Pile / pier, ground improvement (stone columns, preloading)
Rock within 3 m	Footing on rock; rock bolt / anchor if required
Fill / loose sand liquefiable	Compaction, piles to firm layer, remove/replace

5. Preparation of the report - step by step

- Collect all field logs and lab data; check and reconcile (logs vs sample vs test results).
- Group data layer-wise, prepare idealised soil profile and design values (use average/ lower bound).
- Draw BH location plan, subsoil sections, graphs (N vs depth, grain-size, consolidation curves).

- Carry out analysis for foundation type as above (with factor of safety) and check settlement.
- Write the recommendations with clear statements (depth of footing from which G.L., pressure, precautions).
- Review by senior, final printing (colour photographs, drawings at legible scale), signing.

PART B - FIELD INSTRUMENTATION

Geotechnical instrumentation is used to measure pore pressure, total stress, load, strain, settlement, lateral movement - to (i) verify design assumptions, (ii) control construction (rate of filling, staged loading), (iii) warn of failure, and (iv) improve future designs (observational method, Peck).

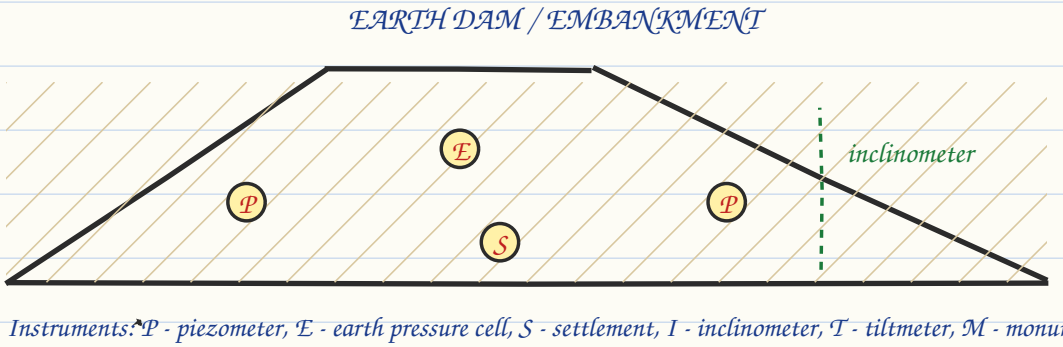
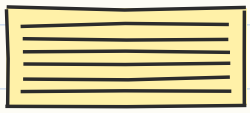


Fig 5.2 Instrumentation of an earth dam

Quantity to measure	Instrument
Pore water pressure / ground water	Piezometers (standpipe, pneumatic, vibrating wire)
Total earth pressure	Pressure cells (earth pressure cells)
Strain in steel / concrete	Strain gauges
Lateral movement	Inclinometer, tiltmeter
Settlement, vertical movement	Settlement gauge, extensometer, level survey
Load in anchors, struts, piles	Load cells, proving rings
Displacement, deformation	Dial gauge, LVDT, crack meter, tape extensometer

6. Strain gauges

Strain gauge (foil type):



grid of Constantan wire

GF = (DR/R)/e (gauge factor ~ 2.0)

e = DL/L strain; stress s = E e

Types: electrical resistance (bonded, foil, weldable)

vibrating wire: e = K(f2 - f0^2)

Quarter / half / full Wheatstone bridge

Fig 5.3 Electrical resistance strain gauge

- Principle: change in length -> change in electrical resistance of the conductor: DR/R = GF x e.
- Bonded foil gauges are glued onto steel/concrete; connected in Wheatstone bridge (temperature compensation by dummy gauge). Reading unit = strain indicator in microstrain.
- Vibrating wire strain gauge: wire in tension; plucked; frequency f measures strain: e = K(f2 - f0^2). Stable for years; used in dams, piles, tunnel linings.
- Embedment gauges (in concrete) and spot weldable gauges (on steel / pile / strut).
- Applications: stresses in pile/ struts / tunnel lining, load tests on piles, rock bolts, retaining wall reinforcement.

Type	Merits	Demerits
Electrical resistance	Cheap, small, good for dynamic	Drift, moisture sensitive, cable length affects
Vibrating wire	Long-term stable, long cables	Not for fast dynamic events, costlier
Mechanical (Demec)	No electronics	Manual, lower accuracy

7. Piezometers

Measure pore water pressure (u) - essential for effective stress analysis and stability of embankments, dams, slopes and excavations.

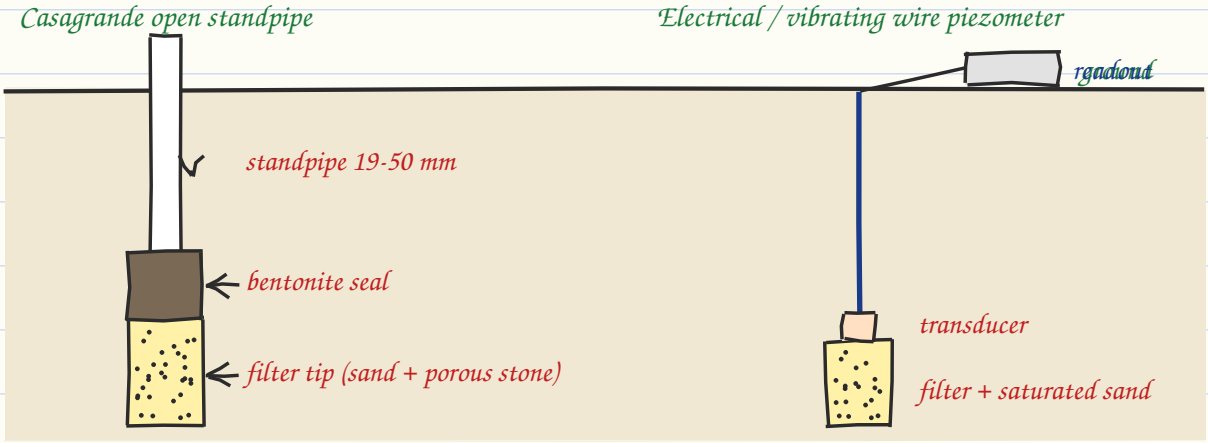


Fig 5.4 Standpipe (Casagrande) and electrical piezometer

Type	Working	Remarks
Open standpipe (Casagrande)	Porous tip connected to a vertical PVC pipe; water level in pipe = piezometric head (dip meter)	Cheap, simple, reliable; slow response (time lag) in clay; max depth limited, freezing issues
Hydraulic (twin-tube)	Two de-aired tubes lead to gauge/panel	Good for fills; needs de-airing
Pneumatic	Diaphragm closes at tip; gas pressure applied until balance - open vent	No lag, remote reading; for earth dams
Vibrating wire	Diaphragm + wire frequency	Excellent long-term, automatic data logging
Electrical (strain-gauge)	Pressure transducer	Quick, but drift

- Installation: borehole -> sand pocket around filter tip (0.3-0.5 m) -> bentonite plug above -> grout/ backfill. Test response (falling-head) and saturate the tip.
- Hydrostatic head $h = u/\gamma_w$; piezometric level compared with elevation gives flow direction. Hvorslev time-lag $T = A/(F\,k)$; where F shape factor.
- Use in clay with slow response: choose diaphragm types.

8. Pressure cells (earth pressure cells)

Measure total stress in soil (or contact pressure between soil and structure: retaining walls, tunnel lining, bridge abutments, dam core). Effective stress = total - pore pressure (from piezometer).

- Construction: two circular stainless steel plates welded at the edge, space between them filled with de-aired oil/mercury; pressure measured by a pneumatic or vibrating-wire transducer (diaphragm type).
- Types: (i) Embedded (total pressure) cells - in soil mass, (ii) Contact cells - flush on structure face, (iii) Hydraulic, pneumatic, vibrating wire etc.
- Installation: cell placed in thin sand/fine soil cushion, orientation noted; stiffness of cell should be similar to soil to avoid arching / over-registration; aspect ratio (diameter: thickness) at least 10 : 1.
- Errors: stress concentration, temperature, installation disturbance, cell/soil stiffness ratio; calibrate in the lab.

9. Inclinometers

Measure lateral (horizontal) deformation vs depth - slopes, landslides, retaining walls, diaphragm walls, embankment toe, piles. Also called slope indicator.

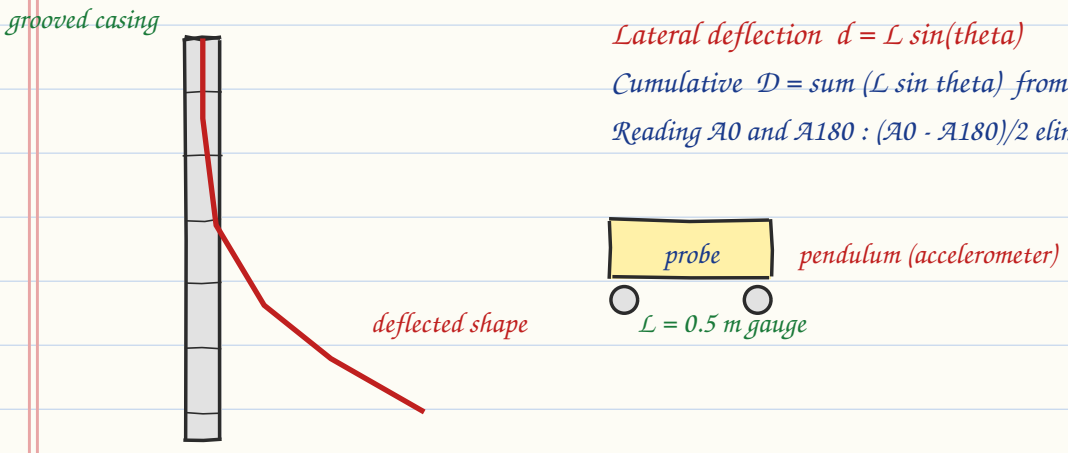


Fig 5.5 Inclinometer casing and probe

- Components: grooved plastic/aluminium casing installed in a borehole with grout, the bottom fixed in stable stratum; probe (torpedo) with two wheels and pendulum/MEMS accelerometer (servo), cable with depth marks, and readout.
- Procedure: lower probe to bottom, take readings at every 0.5 m going up (A0 and A180 face), repeat in perpendicular (B-axis). Compare with the initial reading (baseline).
- Tilt q gives lateral deflection per gauge length: $d = L \sin q$; cumulative deflection at depth z : $D = S\, L \sin q$ (from bottom up).

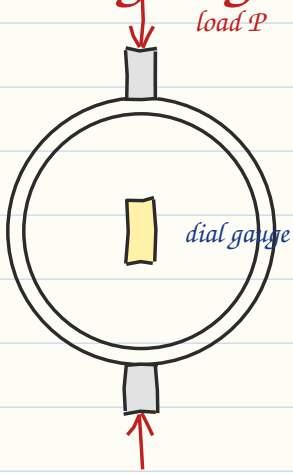
- Output: deflection vs depth, rate of movement, shear zone (sudden change of slope) - slip surface depth.

> Example. Gauge length 0.5 m; readings sin q at successive 0.5 m intervals (from base): 0, 0.002, 0.004, 0.010, 0.016. Deflection at top = 0.5 × (0 + 0.002 + 0.004 + 0.010 + 0.016) = 0.5 × 0.032 = 0.016 m = 16 mm. The largest increments occur between the 3rd-5th points -> shear zone.

Tiltmeter

Measures rotation of a structure/ground (arc-seconds): used for monitoring wall rotation, building tilt, dam crest, adjacent structures during tunnelling.

10. Proving ring



Proving ring: elastic steel ring + dial gauge

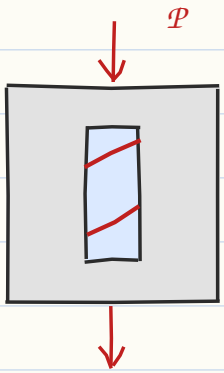
- Load P = K × (dial reading - zero)
- K = calibration constant (N per division)
- Capacity 5 kN - 2000 kN
- Used for CBR, UCS, triaxial, vane
- Calibrated against standard dead weights
- Accuracy about 0.1 - 1 %

Fig 5.6 Proving ring

A precisely machined steel ring with a dial gauge (or LVDT) across its internal diameter. When load is applied along the diameter, the ring deflects elastically and the dial reading is proportional to load. Load P = K × (dial reading) where K is the ring constant from calibration (N/div.).

- Calibration: load with standard weights / compression machine; plot load vs dial reading; check linearity; re-calibrate every year.
- Used in laboratory (UCS, CBR, triaxial, direct shear machines) and also in field for measuring jack loads in plate load tests, pile load tests.
- Advantages: simple, robust, no power; Limitations: manual reading, temperature effect, sensitive to shock.

11. Load cells



Strain gauge load cell

- Steel column with 4 gauges (2 axial, 2 transverse)
- Wheatstone bridge: Vout/Vin ~ GF x e (full bridge)
- Hydraulic load cell: fluid pressure in diaphragm
- Vibrating wire load cell: P = k (f0^2 - f^2)
- Uses: anchor / strut load, pile test, tie rods
- Capacity 10 kN to 10 MN

Fig 5.7 Load cells

Type	Principle	Application
Strain gauge (electrical)	Gauges on elastic steel member; output in mV	Tie-back anchors, struts, pile test
Hydraulic	Load -> pressure in fluid in diaphragm, read by Bourdon gauge/transducer	Heavy loads, jack monitoring
Vibrating wire	Wire frequency in steel body	Long-term monitoring, rock anchors, ground anchors
Pneumatic	Air pressure balances load	Hazardous areas (no electricity)
Photo-elastic	Stress-induced optical change	Rare / research

- Capacity: 10 kN to 10 MN; accuracy 0.5 - 1 % of full scale; compensated for temperature; centrally loaded with bearing plates.
- Calibrate with a testing machine; protect from water (waterproof sealing).

12. Displacement gauges

Gauge	Use
Dial gauge (mechanical)	Settlement in plate load / consolidation tests (0.01 mm resolution, range 25 - 50 mm)
LVDT (linear variable differential transformer)	Electrical displacement transducer; core moves in coils; automatic data acquisition; excellent for lab & field
Extensometer (rod / tape / borehole multi-point)	Relative movement along a borehole; tunnel convergence; rock mass deformation

Gauge	Use
Settlement gauge (magnet-extensometer, hydrostatic profile gauge, plate)	Vertical settlement of fills/embankments at various depth
Crack meters / joint meters	Opening of cracks and joints in structures/ rock
Surveying (total station, GNSS, levelling)	Surface monuments - absolute movement
Fibre optic & Tilt sensors	Distributed strain / deformation

Selection rules for any instrument: (1) Why measure? Define the questions first; (2) predict expected values; (3) choose reliable, accurate, robust instrument with required range; (4) plan calibration, installation, protection and baseline readings; (5) plan data collection, logging, thresholds & action levels; (6) review data continuously.

13. Summary of instruments

Instrument	Measures	Unit	Typical site
Strain gauge	Strain in steel/concrete	microstrain	Piles, struts, tunnel
Piezometer	Pore water pressure	kPa / m head	Dam, slopes, excavation
Earth pressure cell	Total stress	kPa	Retaining wall, embankment
Inclinometer	Lateral deflection	mm	Landslide, diaphragm wall
Proving ring	Load (force)	kN	Lab/field loading tests
Load cell	Load in anchor/strut	kN / MN	Anchors, piles
Displacement gauge	Movement	mm	Settlement, crack

14. Important questions

- What are the contents of a soil exploration report? Explain each briefly.

- Write about the identification and calculations needed in a soil report. Include an example for safe bearing capacity.
- Explain the working of different types of piezometers. Which is best for clay?
- Write short notes on strain gauges and vibrating wire gauges.
- Describe the inclinometer and how lateral displacement is computed.
- Explain the proving ring and its calibration.
- Write short notes on load cells and displacement gauges.
- Why is instrumentation needed in an earth dam? Name suitable instruments.